

EE104T: Network Theory

Assignment-1

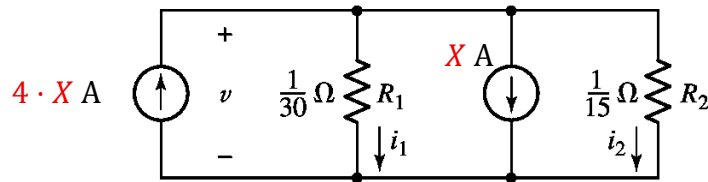
19th Jan' 2026

Total marks - 40

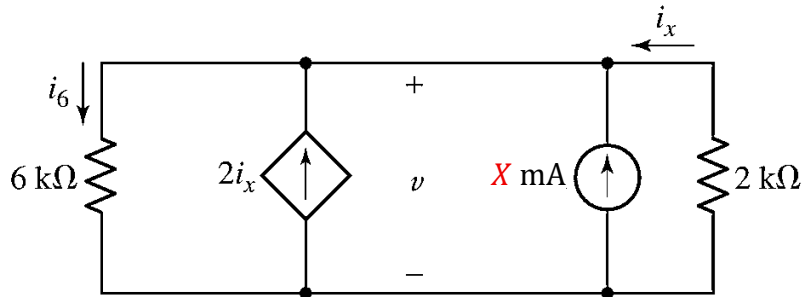
Submission deadline – 27th Jan' 2026**Important Note:**

- Many of the problems have value given by X . The value of X corresponding to your roll number is given on the last page (**Page-6**).
- Answers corresponding to the assigned value of X will only be considered.
- Provide your answers in the .txt file which is attached as a template. The answers should be rounded to the second decimal place.

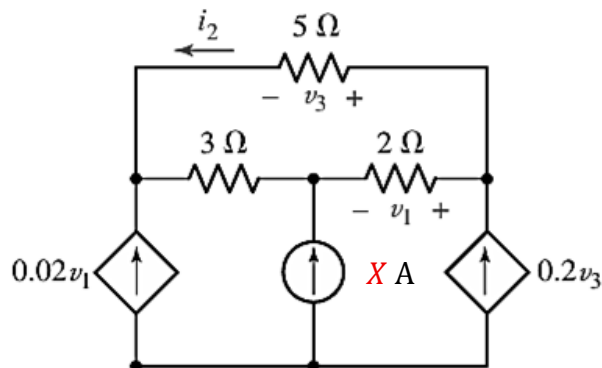
- Determine the currents i_1 and i_2 in the following circuit. Find out the value of power absorbed by resistor R_1 . **(6-mark)**



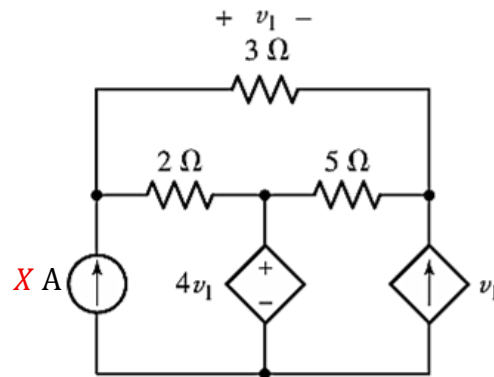
- Determine the value of v and the power supplied by the independent current source in the following circuit. **(4-mark)**



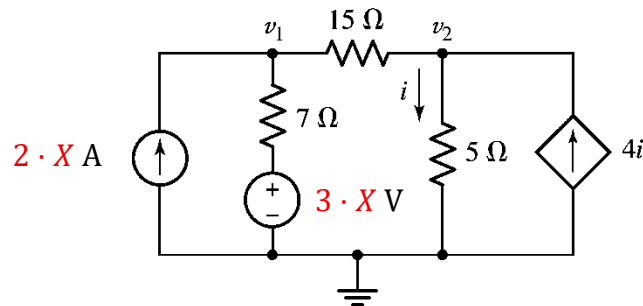
- Determine the current i_2 in the following circuits. **(3-mark)**



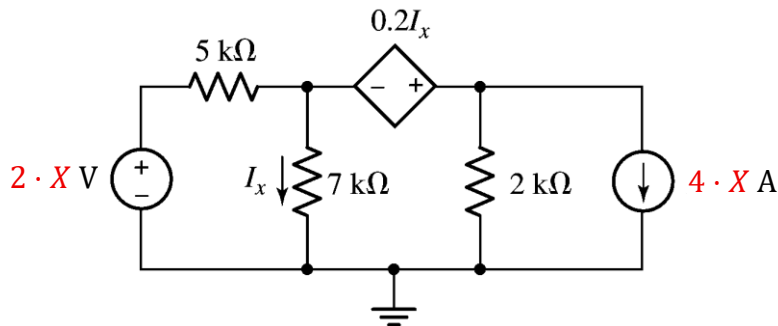
4. Determine the voltage labeled v_1 across the 3Ω resistor. **(3-mark)**



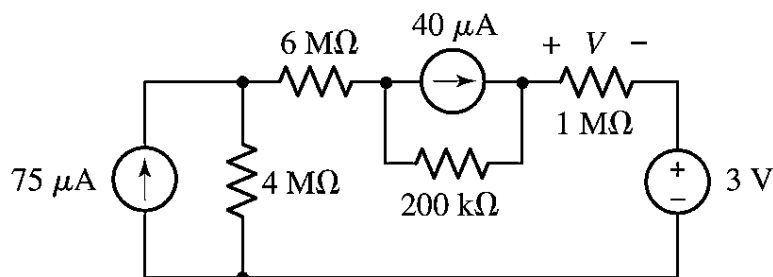
5. For the following circuit, use superposition to obtain the voltages v_1 and v_2 . **(4-mark)**



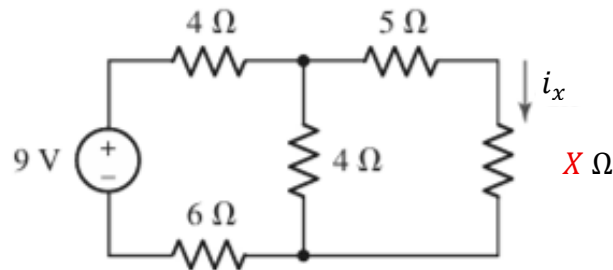
6. Use superposition principles to obtain a value for the current I_x as labeled in the following figure. **(3-mark)**



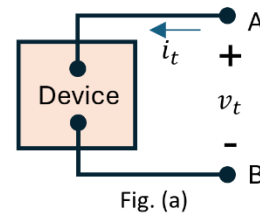
7. For the following circuit, compute the voltage V across the $1\text{ M}\Omega$ resistor using source transformation technique. **(3-mark)**



8. Use Thevenin's theorem to find the current through the $X \Omega$ resistor in the following circuit. **(2-mark)**



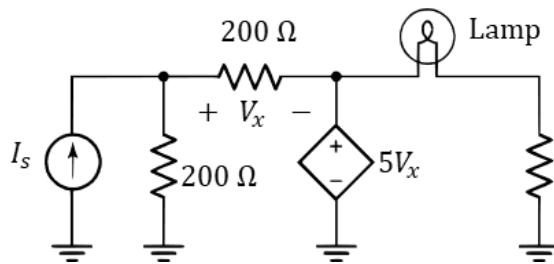
9. The voltage and current are measured at the terminals of the device illustrated in Fig. (a), and the values of v_t and i_t are tabulated in Fig. (b). Obtain the Thevenin resistance (R_{th}) and Thevenin Voltage (V_{th}) for this device. **(6-mark)**



v_t (V)	i_t (A)
80	$1 \cdot X$
70	$2 \cdot X$
60	$3 \cdot X$

Fig. (b)

10. The load resistor shown in the figure can safely dissipate up to 1 W before overheating and bursting into flame. The lamp can be treated as a 10.6Ω resistor if less than 1 A current flows through it and a 15Ω resistor if more than 1 A flows through it. Determine the maximum permissible value of I_s ? **(6-mark)**

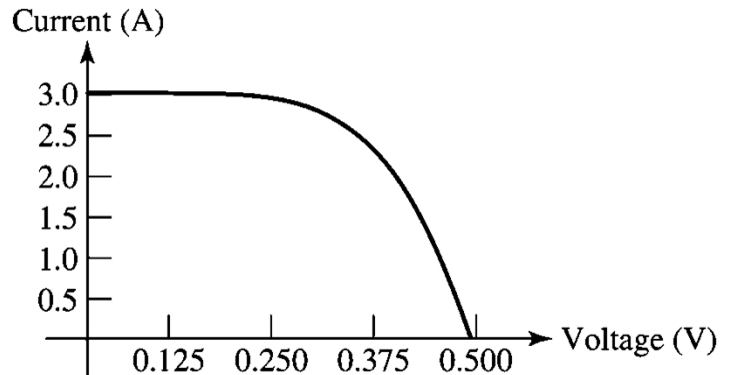


Load-resistor

$$R_L = 1000 + X \Omega$$

11. The current-voltage characteristic of a silicon solar cell exposed to direct sunlight at is given in the following Figure. It is obtained by placing different-sized resistors across the two terminals of the device and measuring the resulting currents and voltages. **(Practice problem - Ungraded)**

- What is the value of the short-circuit current?
- What is the value of the voltage at open circuit?
- Estimate the maximum power that can be obtained from the device.



12. A certain red LED has a maximum current rating of 35 mA, and if this value is exceeded, overheating and catastrophic failure will result. The resistance of the LED is a nonlinear function of its current, but the manufacturer warrants a minimum resistance of $47\ \Omega$ and a maximum resistance of $117\ \Omega$. Only 9 V batteries are available to power the LED. Design a suitable circuit to deliver the maximum power possible to the LED without damaging it. **(Practice problem - Ungraded)**

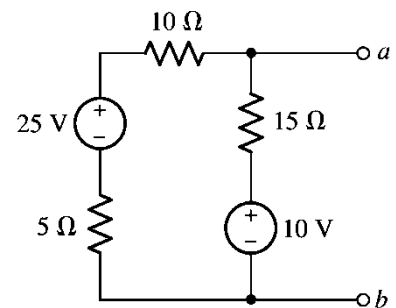
Use only combinations of the standard resistor values given below:

Standard 5 Percent Tolerance Resistor Values

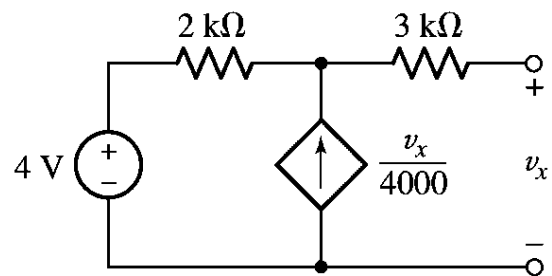
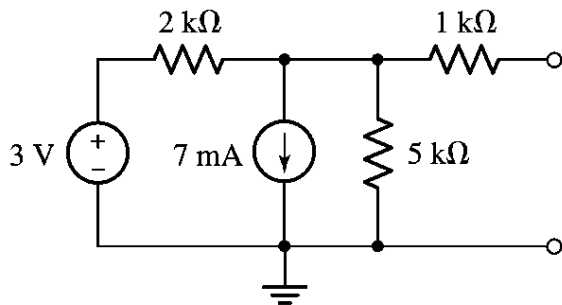
1.0	1.1	1.2	1.3	1.5	1.6	1.8	2.0	2.2	2.4	2.7	3.0	3.3	3.6	3.9	4.3	4.7	5.1	5.6	6.2	6.8	7.5	8.2	9.1	Ω
10.	11.	12.	13.	15.	16.	18.	20.	22.	24.	27.	30.	33.	36.	39.	43.	47.	51.	56.	62.	68.	75.	82.	91.	Ω
100	110	120	130	150	160	180	200	220	240	270	300	330	360	390	430	470	510	560	620	680	750	820	910	Ω
1.0	1.1	1.2	1.3	1.5	1.6	1.8	2.0	2.2	2.4	2.7	3.0	3.3	3.6	3.9	4.3	4.7	5.1	5.6	6.2	6.8	7.5	8.2	9.1	k Ω
10.	11.	12.	13.	15.	16.	18.	20.	22.	24.	27.	30.	33.	36.	39.	43.	47.	51.	56.	62.	68.	75.	82.	91.	k Ω
100	110	120	130	150	160	180	200	220	240	270	300	330	360	390	430	470	510	560	620	680	750	820	910	k Ω
1.0	1.1	1.2	1.3	1.5	1.6	1.8	2.0	2.2	2.4	2.7	3.0	3.3	3.6	3.9	4.3	4.7	5.1	5.6	6.2	6.8	7.5	8.2	9.1	M Ω

13. It is known that a load resistor of some type will be connected between terminals a and b of the network shown in the following figure. **(Practice problem - Ungraded)**

- Change the value of the 25 V source such that both voltage sources contribute equally to the power delivered to the load resistor, assuming its value is chosen such that it absorbs maximum power.
- Calculate the value of the load resistor.

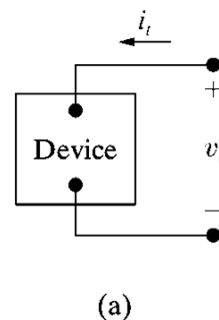


14. Determine the Thevenin and Norton equivalent of the circuits shown below. (**Practice problem - Ungraded**)



15. The voltage and current were measured at the terminals of the device shown in Fig.(a). The results are tabulated in Fig.(b). (**Practice problem - Ungraded**)

- Construct a circuit model for this device using an ideal current source in parallel with a resistor.
- Use the model to predict the amount of power the device will deliver to a resistor.



v_t (V)	i_t (A)
100	0
120	4
140	8
160	12
180	16

(b)

Roll No	X
EC25BT001	88
EC25BT002	98
EC25BT003	74
EC25BT004	109
EC25BT006	106
EC25BT007	27
EC25BT009	85
EC25BT010	32
EC25BT011	87
EC25BT012	119
EC25BT013	58
EC25BT014	53
EC25BT015	21
EC25BT016	70
EC25BT017	67
EC25BT018	95
EC25BT019	90
EC25BT020	108
EC25BT021	53
EC25BT022	31
EC25BT023	108
EC25BT024	105
EC25BT025	100
EC25BT026	81
EC25BT027	116
EC25BT028	91
EC25BT029	81
EC25BT030	45
EC25BT031	84
EC25BT033	101
EC25BT034	62
EC25BT035	51
EC25BT036	92
EC25BT037	73
EC25BT038	88
EC25BT039	115
EC25BT040	102
EC25BT041	91
EC25BT042	105
EC25BT043	31
EC25BT044	82

Roll No	X
EE25BT002	92
EE25BT003	60
EE25BT004	73
EE25BT005	53
EE25BT006	56
EE25BT007	67
EE25BT008	91
EE25BT009	24
EE25BT010	53
EE25BT011	42
EE25BT012	69
EE25BT013	69
EE25BT014	46
EE25BT015	41
EE25BT018	36
EE25BT020	82
EE25BT021	108
EE25BT022	24
EE25BT023	113
EE25BT024	110
EE25BT025	63
EE25BT026	99
EE25BT028	64
EE25BT029	54

Quiz-1 – EE104T: Network Theory (Jan – Apr 2026)

2nd Feb' 2026

Name: _____, Roll No: _____, Total marks: 20, Duration: 45 min

Q1: The voltage and current relationship of a certain device is obtained experimentally and is shown in Fig. (a). **(4-marks)**

- Obtain the Thevenin's Equivalent Representation of this device.
- Determine the value of v_t that leads to maximum power transfer from this device?

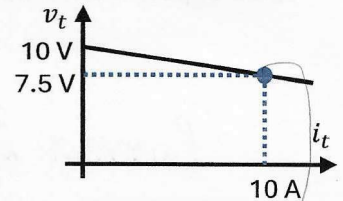
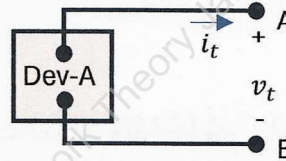
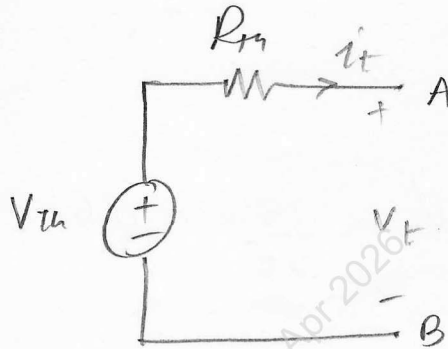


Fig. (a)

(a)



No partial grading

$$V_{th} = V_{AB} \Big|_{i_t=0} = 10 \text{ V}$$

1-mark for correct value of V_{th}

$$V_t = V_{th} - R_{th} \cdot i_t$$

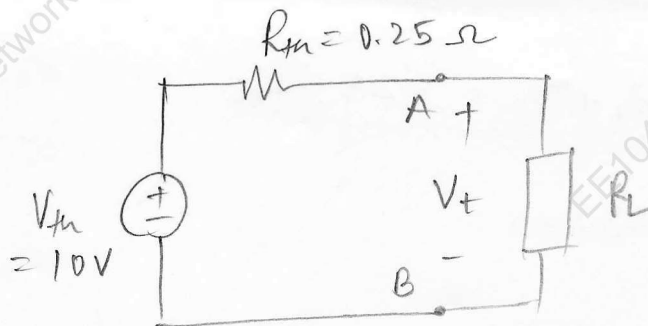
$$7.5 = 10 - R_{th} \cdot 10$$

$$\Rightarrow R_{th} = 0.25 \Omega$$

using this values.

1-mark for correct value of R_{th}

(b)



No partial grading

We know that maximum power is transferred when $R_L = R_{th}$ i.e., $R_L = R_{th} = 0.25 \Omega$

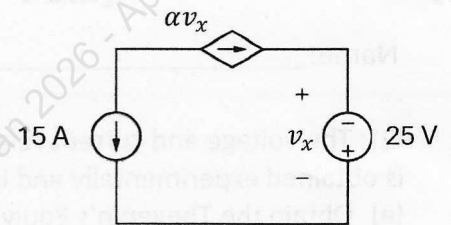
$$\therefore V_t = \frac{V_{th}}{2} = \underline{\underline{5 \text{ V}}}$$

2-marks for correct value of v_t

This value is independent of R_L or R_{th} !

Q2: Determine the value of α that makes this connection valid.

Mention the unit of α . For the obtained value of α , determine the value of power absorbed by 15 A current source. **(3-marks)**



Current in a series network is same

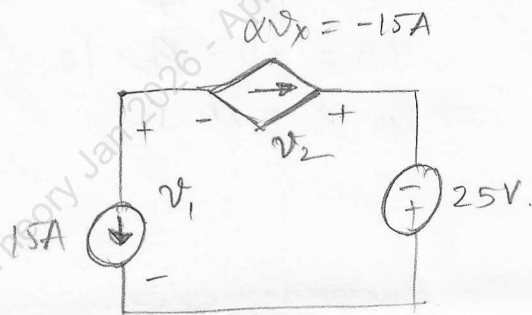
$$\therefore 15 \text{ A} = -\alpha v_x$$

also $v_x = -25 \text{ V}$

$$\therefore 15 \text{ A} = \alpha 25 \text{ V} \Rightarrow \alpha = \frac{15}{25} = 0.6$$

No partial grading

2-marks for correct value of alpha



Here, we find that $+v_1 + v_2 = -25$,

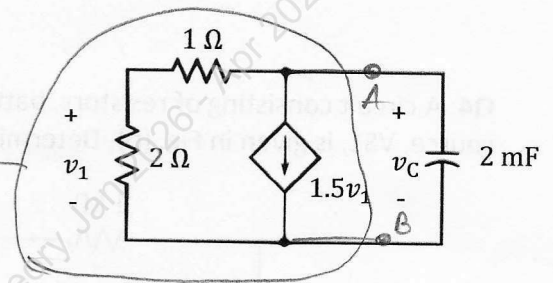
$\therefore v_1$ has infinite solutions

1-mark for power-absorbed justification

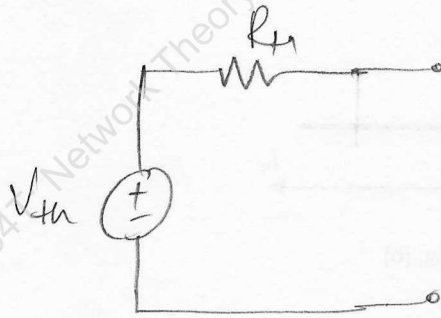
$\therefore$ Power-absorbed by 15 A source = $15 \text{ A} \cdot v_1$ has several solutions

Q3: In the circuit shown here, determine the expression for voltage $v_C(t)$ for $t > 0$ if $v_C(0^-) = 10 \text{ V}$.

(4-marks)

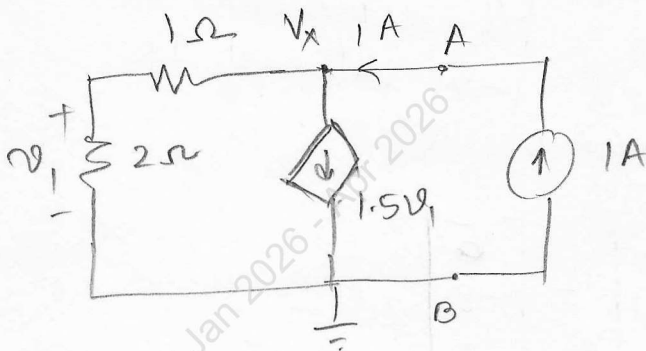


← Thevenin's Equivalent



$V_{th} = 0$ since there are no active sources.

1-mark for correct value of V_{th}



$$1 \text{ A} = 1.5v_1 + \frac{v_x}{3} \quad \text{--- (1)}$$

$$8v_1 = 2 \cdot (1 - 1.5v_1)$$

$$\therefore v_1 = 2 - 3v_1$$

$$\therefore v_1 = 0.5 \text{ V} \quad \text{--- (2)}$$

putting (2) in (1) gives.

$$1 = 1.5(0.5) + \frac{v_x}{3}$$

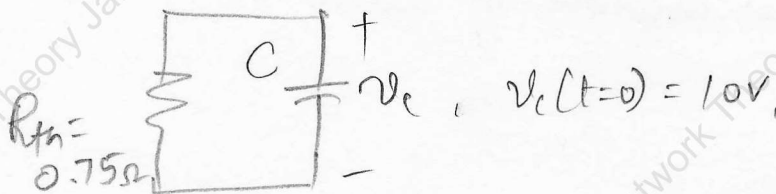
$$v_x = 0.75 \text{ V} = v_{AB}$$

No partial grading

$$\therefore R_{th} = \frac{v_x}{1 \text{ A}} = 0.75 \Omega$$

1-mark for correct value of R_{th}

$\therefore$ The given circuit can be equivalently represented as.

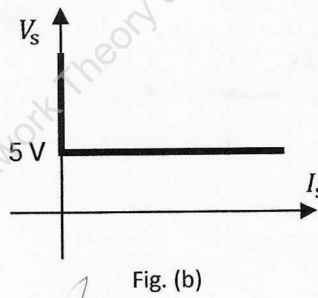
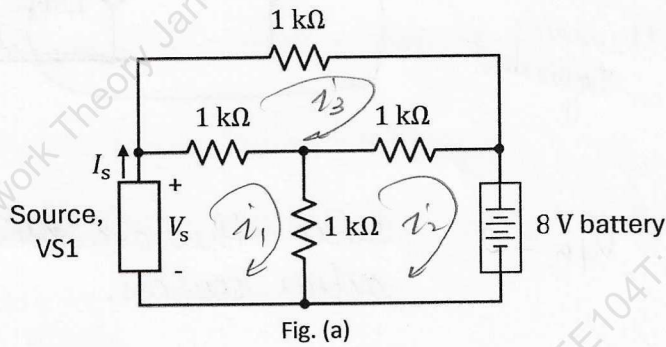


The capacitor discharges exponentially from initial value $\therefore v_C(t) = 10 e^{-t/\tau}$ where $\tau = R_{th} \cdot C$

2-marks for correct expression of v_C

$$\tau = 0.75 \times 2 \times 10^{-3} = 1.5 \text{ ms}$$

Q4: A circuit consisting of resistors, battery and a source is shown in Fig. (a). The terminal-characteristic of the source, VS1, is given in Fig. (b). Determine the value of power-absorbed by the source VS1. (5-marks)



Assume I_s is positive; and called i_1
 $\therefore V_s = 5V$

$$\begin{bmatrix} 2k & -1k & -1k \\ -1k & 2k & -1k \\ -1k & -1k & 2k \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 5 \\ -8 \\ 0 \end{bmatrix}$$

deduct 0.5 marks
for each incorrect
entry in this form

2-marks for procedure: Either standard form of Mesh Analysis, or three correct KVL/KCL equations

The source does not allow
-ve values of I_s .

1-mark for identifying that VS1
can not absorb power

we get that i_1 is negative, means our assumption is
incorrect and $i_1 = 0$

$$\therefore \text{Power absorbed by source } V_{s1} = V_s (-I_s) = 0V$$

2-marks for
correct value of
power-absorbed

Q5: A series RC-circuit shown here is to be solved using numerical approach on a computer. The variable to be solved for is v , voltage across the capacitor.

(a) Obtain the expression for $v(n+1)$ in terms of $v(n)$, time-step, Δt and other circuit parameters.

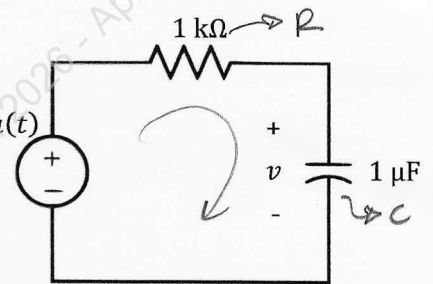
(b) Given that the expression obtained in (a) is to be solved on a computer,

(i) What should be the initial value of v , i.e. $v(n=0)$?

(ii) What should be the value of Δt ?

(4-marks)

[Hint: First obtain the first-order differential equation in terms of v .]



(a) Using KVL we get

$$v_s = iR + v \quad \text{and} \quad i = C \frac{dv}{dt}$$

$$\therefore v_s = RC \frac{dv}{dt} + v$$

$$\therefore \frac{dv}{dt} = -\frac{v}{RC} + \frac{v_s}{RC}$$

1-mark for correct expression of first-order ODE.

$\frac{\Delta v}{\Delta t}$

$$\frac{v(n+1) - v(n)}{\Delta t} = -\frac{1}{RC} \cdot v(n) + \frac{v_s(n)}{RC}$$

$$\therefore v(n+1) = v(n) + \Delta t \left(-\frac{1}{RC} \cdot v(n) + \frac{v_s(n)}{RC} \right)$$

1-mark for this expression

(b) (i) initial value of v , i.e. $v(n=0)$ can be chosen based on the initial condition gives. If the energy stored in this network was zero, then $v(n=0) = 0$ V can be selected as the initial value.

1-mark for correct response

(ii) The time constant of this RC network is $\tau = RC$

$$= 1 \text{ k}\Omega \times 1 \mu\text{F}$$

$$\tau = 1 \text{ ms.}$$

For the approximation $\frac{\Delta v}{\Delta t} \approx \frac{dv}{dt}$ to be accurate,

Δt should be much smaller than 1 ms ,

thumb rule $\Delta t < \frac{\tau}{10}$

1-mark for correct response

EE205: Network Theory

Assignment-2

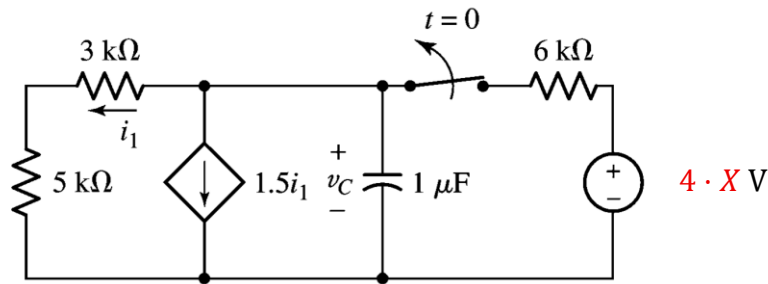
14th Feb'2026

Total marks - 50

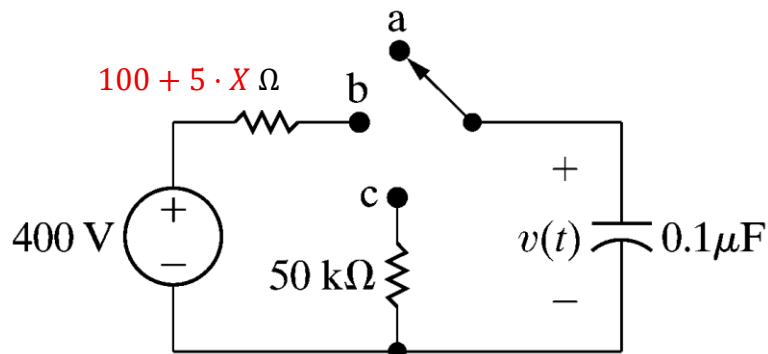
Submission deadline – 23rd Feb' 2026**Important Note:**

- Many of the problems have value given by X . The value of X corresponding to your roll number is given on the last page (**Page-7**).
- Answers corresponding to the assigned value of X will only be considered.
- Provide your answers in the .txt file which is attached as a template. The answers should be rounded to the second decimal place.

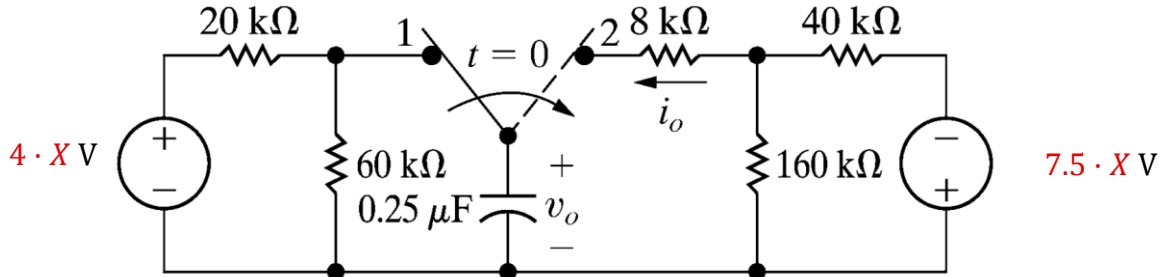
1. For the circuit shown below, determine (a) $v_C(0^-)$ in V; (b) $v_C(0^+)$ in V; (c) the circuit time constant in ms; (d) $v_C(3 \text{ ms})$ in V. **(8-mark)**



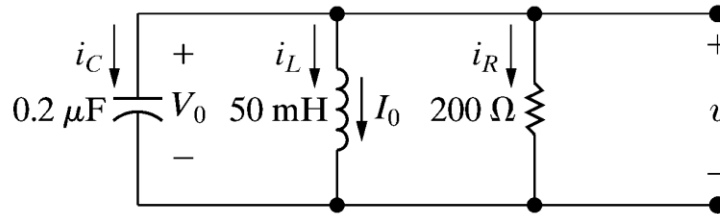
2. The uncharged capacitor in the circuit shown below is initially switched to terminal 'a' of the three-position switch. At $t = 0$, the switch is moved to position 'b', where it remains for 15 ms. After this 15 ms delay, the switch is moved to position 'c', where it remains indefinitely. **(2-mark)**
- (a) Determine the value of t in ms, when the voltage across the capacitor equals 200 V?
- (b) Write the numerical expression for the voltage across the capacitor. (practice problem, ungraded)**
- (c) Plot the capacitor voltage versus time. **(practice problem, ungraded)**



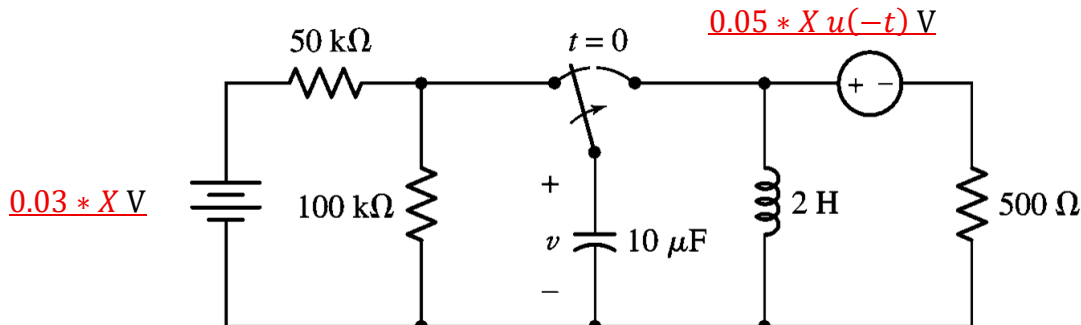
3. The switch in the circuit shown in following figure has been in position '1' for a long time. At $t=0$, the switch moves to position '2'. **(4-mark)**
Find (a) $v_o(t = 1\text{ms})$ in mV and (b) $i_o(t = 1\text{ms})$ in μA .



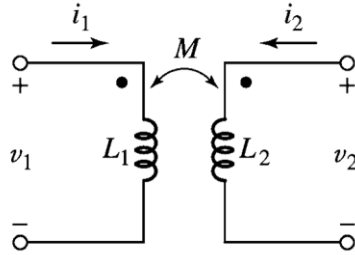
4. For the circuit in following figure, $v(0^+) = 0.12 * X \text{ V}$, and $i_L(0^+) = 30 * X \text{ mA}$. **(4-mark)**
(a) Find the initial value of dv/dt in V/s.
(b) Find the initial current in the capacitor in mA.
(c) Find the expression for $v(t)$. **(practice problem, ungraded)**
(d) Sketch $v(t)$ in the interval $0 \leq t \leq 250 \text{ ms}$. **(practice problem, ungraded)**
(e) Derive the expressions that describe the three branch currents i_R , i_L , and i_C for $t \geq 0$. **(practice problem, ungraded)**



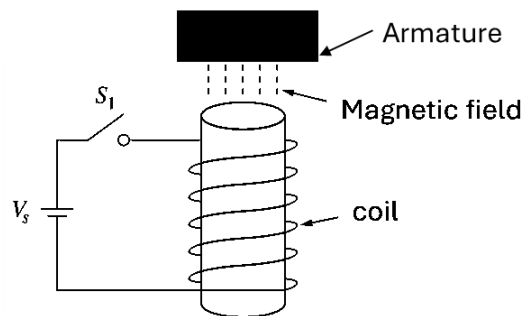
5. The switch in the following circuit has been in the left position for a long time; it is moved to the right at $t = 0$. Find (a) dv/dt at $t = 0^+$ in V/s; (b) v at $t = 1 \text{ ms}$ in V; (c) time-instant t_0 (in ms), the first value of t greater than zero at which $v = 0$. **(6-mark)**



6. In following Figure, let $L_1=0.4\text{H}$, $L_2=2.5\text{ H}$, $k=\frac{X}{100}$, and $i_1 = 4i_2 = 20\cos(500t - 20^\circ)\text{ mA}$. Determine both $v_1(0)$ in V and the total energy stored (in joule) in the system at $t = 0$. **(4-mark)**

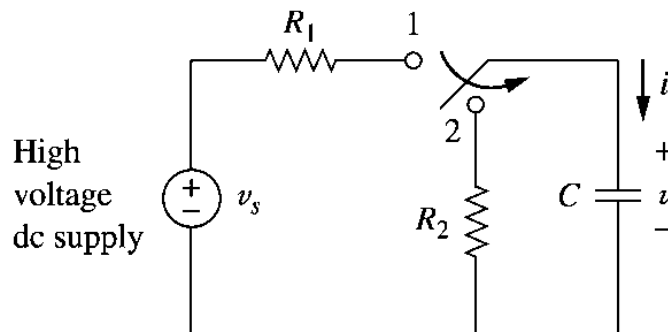


7. An electromechanical relay consists of electromagnet-coil and an armature, as shown below. When the coil is excited with some current, the armature is pulled towards the electromagnet. The current required to pull the armature of a certain relay is $X\text{ mA}$. This relay is operated by a 12-V battery. If the coil has a resistance of $150\ \Omega$ and an inductance of 30 mH , calculate the time (in ms) it takes for the armature to be pulled. **(4-marks)** (Assume that the inductance value does not change with armature position)

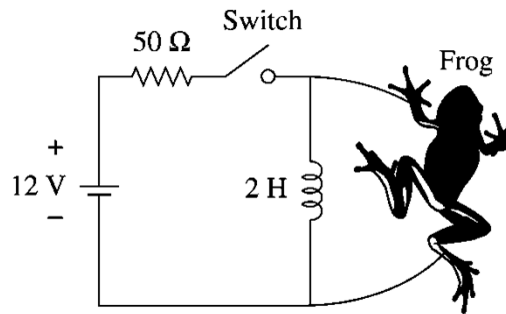


8. An electronic flash application exploits the ability of the capacitor to oppose any abrupt change in voltage. Following figure shows a simplified circuit. It consists essentially of a high-voltage dc supply, a current-limiting large resistor R_1 and a capacitor C in parallel with the flashlamp of low resistance R_2 .

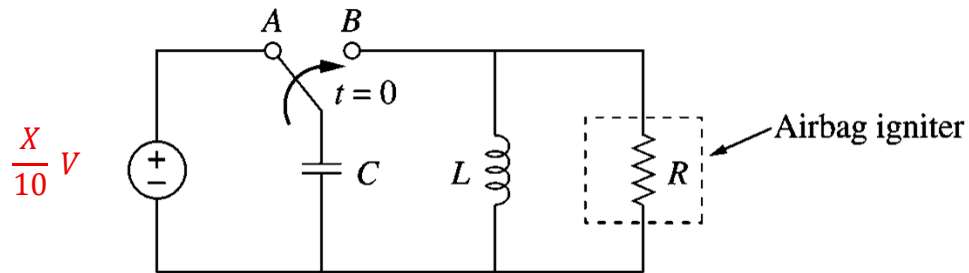
An electronic flashgun has a current-limiting resistor $6\text{ k}\Omega$ and $2000\ \mu\text{F}$ electrolytic capacitor charged to $2.4 * X\text{ V}$. If the lamp resistance is $12\ \Omega$, Determine: (a) the peak charging current (in mA), (b) the time required for the capacitor to fully charge (in ms), (c) the peak discharging current (in mA), (d) the total energy stored in the capacitor (in joule). **(8-marks)**



9. The circuit in the following figure is used by a biology student to study “frog kick.” She noticed that the frog kicked a little when the switch was closed but kicked violently for $X/100$ second when the switch was opened. Model the frog as a resistor and determine the value of this resistance (in ohm). Assume that it takes 10 mA for the frog to kick violently. **(5-marks)**



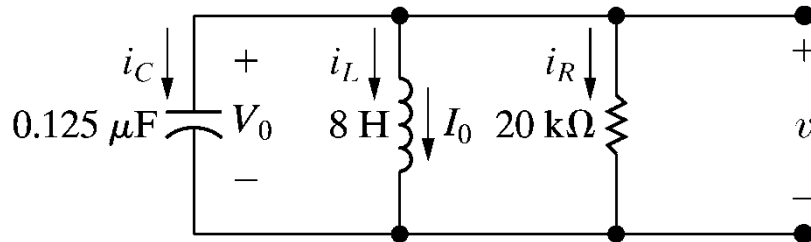
10. An automobile airbag igniter is modeled by the circuit shown below. Determine the time (in ms) it takes the voltage across the igniter to reach its first peak after switching from A to B. Let $R = 3\ \Omega$, $C = 1/30\text{ F}$, and $L = 60\text{ mH}$. **(5-marks)**



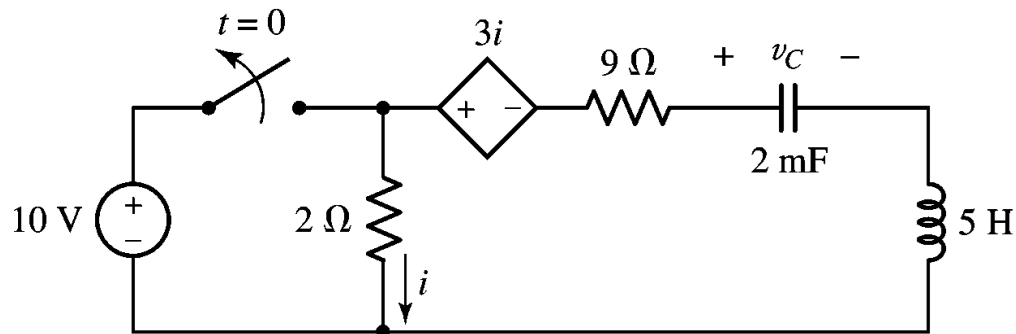
Practice Problems:

11. In the circuit shown below, $V_0 = 0$, and $I_0 = -12.25$ mA. **(3-mark)**

- Calculate the roots of the characteristic equation.
- Calculate v and dv/dt at $t = 0^+$.
- Calculate the voltage response for $t \geq 0$.

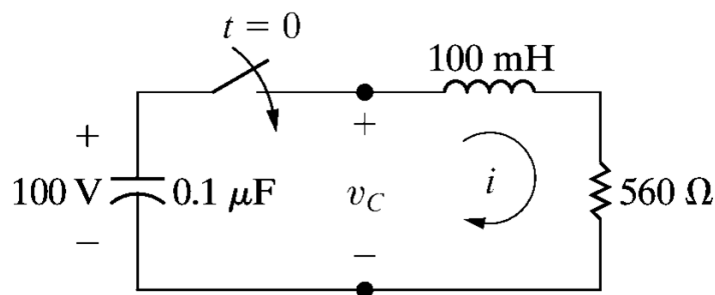


12. Find an expression for $v_C(t)$ in the circuit shown below, valid for $t > 0$. **(3-mark)**

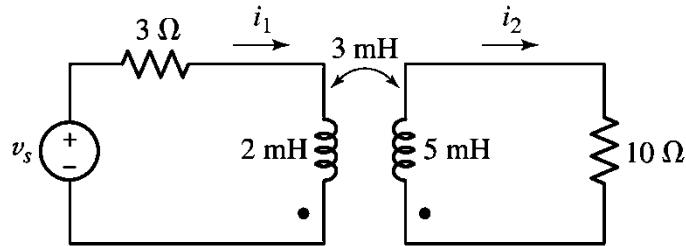


13. The $0.1 \mu\text{F}$ capacitor in the circuit shown below is charged to 100 V. At $t = 0$, the capacitor is discharged through a series combination of a 100 mH inductor and a 560Ω resistor. **(3-mark)**

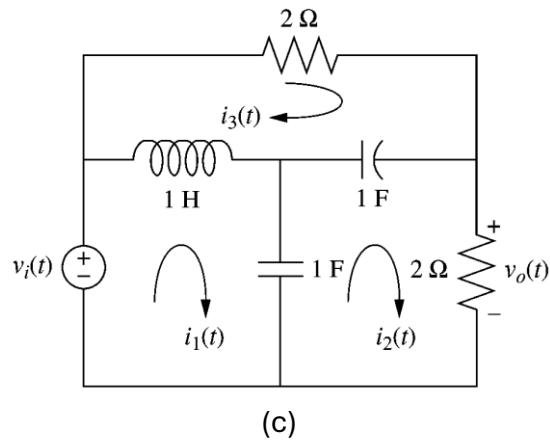
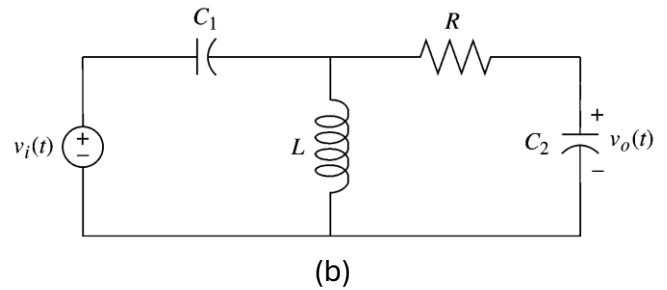
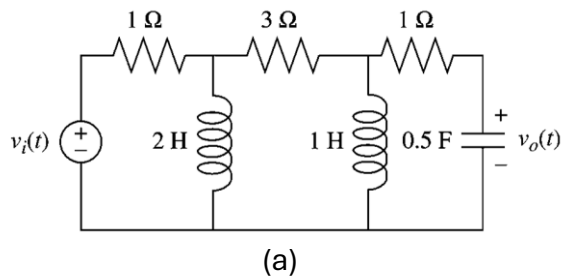
- Find $i(t)$ for $t \geq 0$.
- Find $v_C(t)$ for $t \geq 0$.



14. For the circuit given below, write appropriate mesh equations for the left mesh and the right mesh if $v_s = 20e^{-1000t}$ V. **(4-mark)**



15. Represent the electrical networks shown below in state-space, where $v_o(t)$ is the output and $v_i(t)$ is the input. **(9-mark)**



Roll No	X
EC25BT001	88
EC25BT002	98
EC25BT003	74
EC25BT004	109
EC25BT006	106
EC25BT007	27
EC25BT009	85
EC25BT010	32
EC25BT011	87
EC25BT012	119
EC25BT013	58
EC25BT014	53
EC25BT015	21
EC25BT016	70
EC25BT017	67
EC25BT018	95
EC25BT019	90
EC25BT020	108
EC25BT021	53
EC25BT022	31
EC25BT023	108
EC25BT024	105
EC25BT025	100
EC25BT026	81
EC25BT027	116
EC25BT028	91
EC25BT029	81
EC25BT030	45
EC25BT031	84
EC25BT033	101
EC25BT034	62
EC25BT035	51
EC25BT036	92
EC25BT037	73
EC25BT038	88
EC25BT039	115
EC25BT040	102
EC25BT041	91
EC25BT042	105
EC25BT043	31
EC25BT044	82

Roll No	X
EE25BT002	92
EE25BT003	60
EE25BT004	73
EE25BT005	53
EE25BT006	56
EE25BT007	67
EE25BT008	91
EE25BT009	24
EE25BT010	53
EE25BT011	42
EE25BT012	69
EE25BT013	69
EE25BT014	46
EE25BT015	41
EE25BT018	36
EE25BT020	82
EE25BT021	108
EE25BT022	24
EE25BT023	113
EE25BT024	110
EE25BT025	63
EE25BT026	99
EE25BT028	64
EE25BT029	54

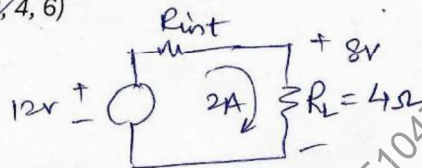
EE104T: Network Theory

Instructor: Amarkumar Kushwaha

Mid-Semester Examination – Feb 2026

Q1: Fill the third column of the following table selecting the correct option. (10-marks)

- independent 1. The Thévenin equivalent resistance is found by deactivating all _____ sources. (Options: dependent, independent, current-controlled, voltage-controlled)
- 1/RC 2. The natural response of a series-RC circuit is characterized by an exponential term of the form e^{-at} where $a =$ _____. (Options: R/C , $1/RC$, RC , C/R)
- real & distinct 3. For an overdamped second-order circuit, the roots of the characteristic equation are _____. (Options: real and equal, real and distinct, purely imaginary, complex conjugates)
- external 4. In forced response analysis, the steady-state solution depends on the _____ input. (Options: natural, homogeneous, external, initial)
- 0 and 1 5. The coefficient of coupling k between two coils lies between _____. (Options: 0 and 1, 1 and 2, -1 and 1, 0 and ∞)
- positively 6. The dot convention helps determine the _____ of mutually induced voltage. (Options: magnitude, polarity, frequency, resistance)
- zero 7. If a linear resistive network contains only dependent sources and no independent sources, the Thévenin equivalent voltage is always _____. (Options: zero, non-zero, infinite, undefined)
- power 8. In a purely resistive network, the superposition theorem cannot be directly applied to calculate _____. (Options: voltage, current, power)
- zero 9. If the initial energy stored in an RLC circuit is zero and no independent sources are present, the response is _____. (Options: oscillatory, zero, exponential, unstable)
- 2 10. A practical 12 V source delivers 2 A to a 4 Ω load. The internal resistance of this source is _____ Ω . (Options: 1, 2, 4, 6)



$$R_{in} = \frac{4V}{2} = 2\Omega$$

1-mark for each correct response

Q2: Some common electrical devices and their nominal voltages are mentioned below. Obtain the scale/order of their terminal currents. For example, a phone charges at ~2A not at 0.2 A or 20 A. (4-marks)

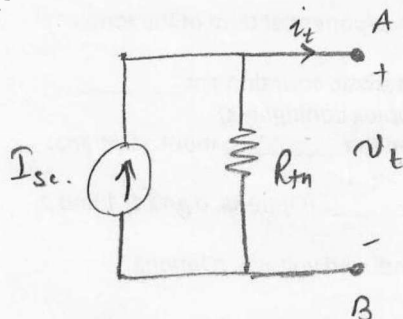
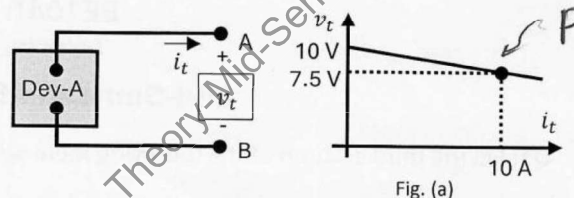
Sr. No.	Electrical device	Nominal operating voltage	Scale/order of Input Current
1	Household Ceiling Fan	230 V	(100 - 200 mA)
2	Electric motor of a four-wheeler EV	400 V	250 A (100 - 300 A)
3	Electric iron	230 V	(2 - 5) A
4	Generators in thermal power-plants	11 kV	8000 A - 12000 A
5	Mobile phone charger	230 V	100 - 300 mA
6	Electric motor of a two-wheeler EV	48 V	80 - 120 A
7	Indian Railways Locomotive	25 kV	100 - 300 A
8	A single LED	2-3 V	5 mA - 50 mA

0.5 marks for each correct response

Power.
30-40W
~180kW
1000W
100MW
65W
5kW
5MW
100MW

Q3: The voltage and current relationship of a certain device is obtained experimentally and is shown in Fig. (a). (4-marks)

- (a) Obtain the Norton's Equivalent Representation of this device.
 (b) Determine the value of i_t that leads to maximum power transfer from this device?



$$i_t = I_{sc} - \frac{v_t}{R_{th}}$$

$$R_{th} = \frac{2.5}{10} = 0.25 \Omega$$

Substituting values corresponding to point 'P'

$$10 = I_{sc} - \frac{7.5}{0.25}$$

$$\Rightarrow I_{sc} = 40 A$$

2-marks for correct values of I_{sc} and R_{th} .

If any of these values are incorrect, deduct 1-mark. If both are incorrect, give 0-marks

(b)

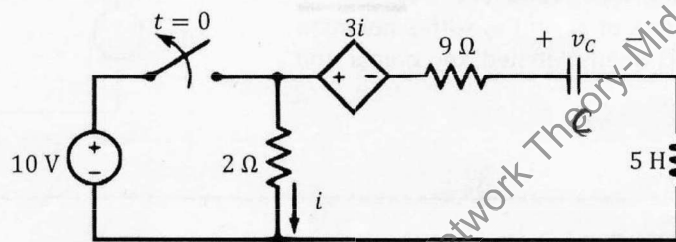
Max-power when $R_{th} = R_{load}$

$\Rightarrow$ current I_{sc} gets equally divided into R_{th} & R_{load}

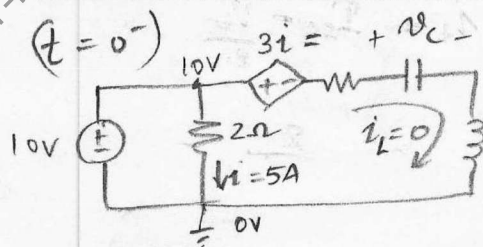
$$\therefore i_t = \frac{I_{sc}}{2} = 20 A$$

2-marks for correct value of i_t . No partial grading.

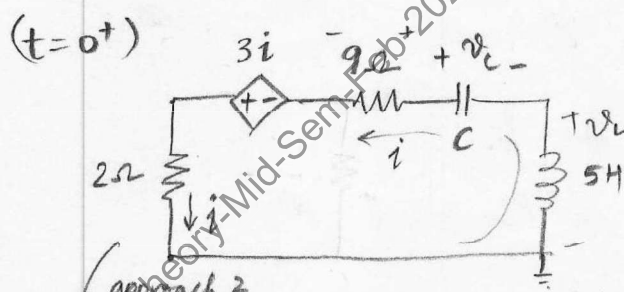
Q4: Obtain the second order ODE that solves for $v_C(t)$ in the following circuit, valid for $t > 0$.
Comment on the nature of the response of v_C . (5-marks)



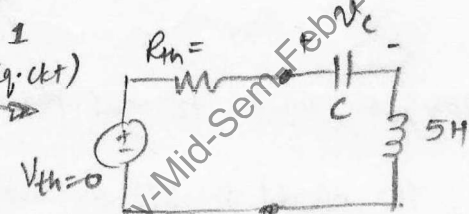
assume the value of capacitor is 'C'.



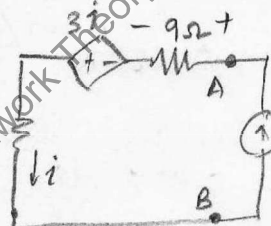
$$i = 5A \\ 3i = 15V \Rightarrow v_C(0^-) = -5V$$



approach 1
(Thevenin eq. dt)



approach 2
(KVL + KCL)



$$\Rightarrow V_{AB} = 2V \\ -3i \\ +9i \\ = 18V$$

$$R_{th} = \frac{V_{AB}}{1A} = 18\Omega$$

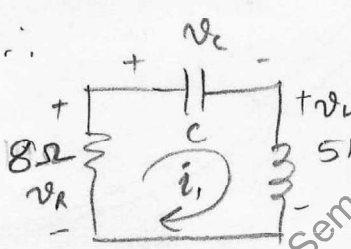
$$2 \cdot i - 3i + 9i - v_C - L \frac{di}{dt} = 0 \\ i = C \frac{dv_C}{dt}$$

2-marks for correct procedure.

$$8 \cdot C \frac{dv_C}{dt} - v_C - 5 \frac{d^2 v_C}{dt^2} = 0$$

$$\therefore 5 \frac{d^2 v_C}{dt^2} - 8C \frac{dv_C}{dt} + v_C = 0$$

2-marks for correct final expression



well known 2nd order series RLC-circuit!

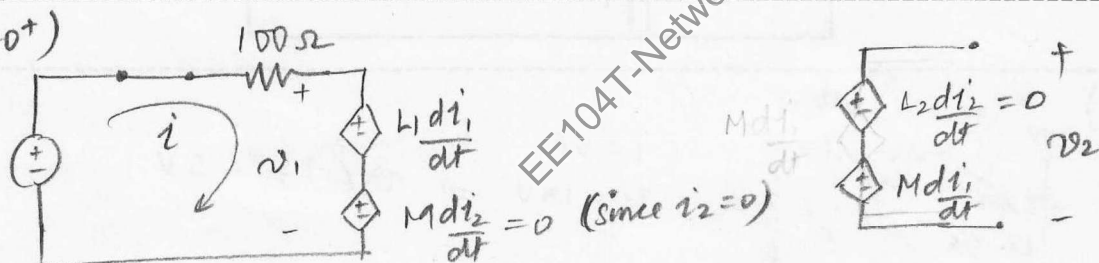
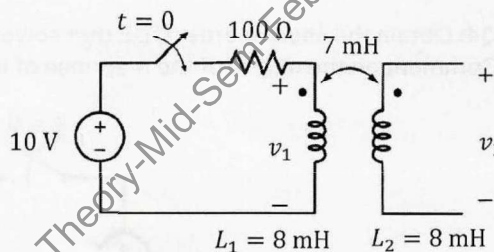
depending on the value of C,
if $\alpha = \frac{R}{2L} = 1.4 > \omega_0 = \frac{1}{\sqrt{LC}}$

- ① response $\rightarrow$ overdamped,
- ② if $\alpha = \omega_0 \rightarrow$ critically damped
- ③ $\alpha < \omega_0 \rightarrow$ under-damped

1-mark for comment on the nature of response. The nature would depend on the value of C.

Q5: For the given coupled inductor network, the switch is moved to closed position at $t = 0$. (6-marks)

- Obtain the expression for v_2 valid for $t \geq 0$.
- Sketch the waveforms of v_1 and v_2 with a common horizontal axis and neatly labeled key points and quantities.



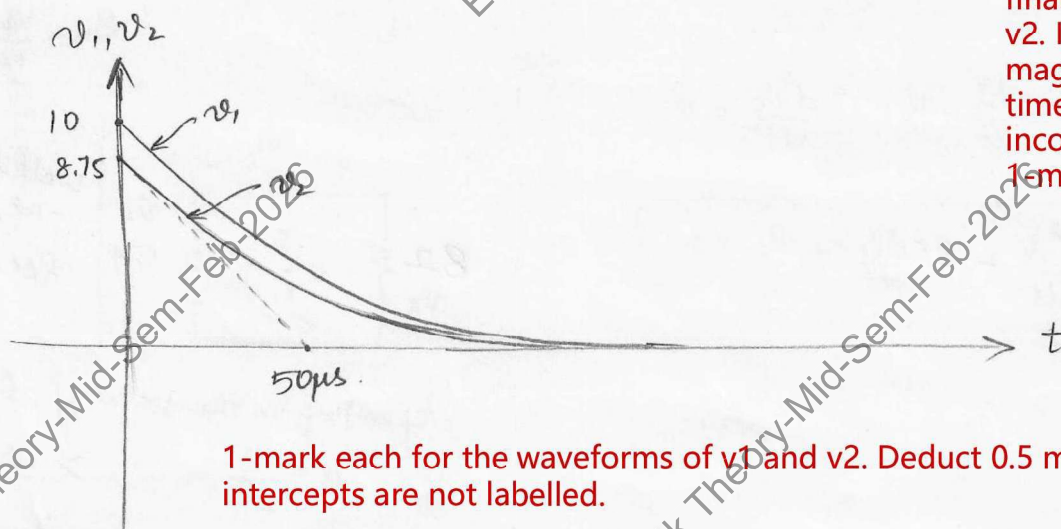
from above : $v_1 = L_1 \frac{di_1}{dt}$ and $v_2 = M \frac{di_1}{dt} = (0.007) \frac{di_1}{dt}$

Now, we need to get the expression for v_1 2-marks for correct procedure.

The circuit on left is simply a series-RL circuit.

$\therefore v_1 = 10 e^{-t/\tau}$ where $\tau = \frac{L}{R} = \frac{0.008}{100} = 50 \mu s$

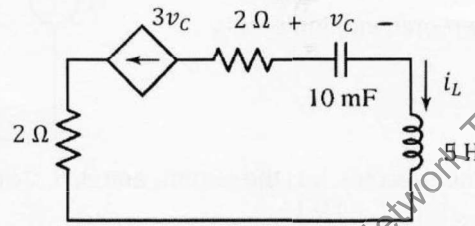
$\therefore v_2 = 0.007 \frac{di_1}{dt} = (0.007) \left(\frac{20}{L_1} \right) = 0.007 \times \frac{10 e^{-t/\tau}}{0.008} = 8.75 e^{-t/\tau}$



1-mark each for the waveforms of v_1 and v_2 . Deduct 0.5 marks if the y-intercepts are not labelled.

2-mark for correct final expression of v_2 . If either magnitude of time-constant is incorrect, deduct 1-mark

Q6: The following network is supposed to be solved for i_L . The initial conditions given are $v_C(0^-) = 10$ V and $i_L(0^-) = 0$ A. Identify the error with this problem formulation. (5-marks)



$3v_C = i_L$ should satisfy at all times since this is a series circuit!

However, $v_C(0^-) = 10\text{V} \Rightarrow 3v_C(0^-) = 30$

But $i_L(0^-)$ is given as 0 A

$\therefore 3v_C \neq i_L$ at $t=0^-$.

The initial conditions are inconsistent.

5-marks for correct response. No partial grading. Multiple solution approaches are possible, grade accordingly.

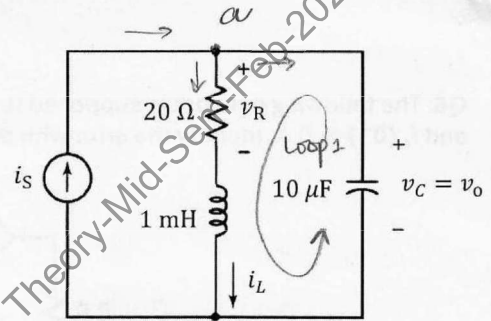
Q7: In the given electrical network, v_o is the output and $i_s(t)$ is the input. The initial conditions for inductor currents and capacitor voltage are: $i_L(0) = 10$ A and $v_o(0) = 200$ V.

(12-marks)

(a) Obtain the complete state-space representation of this network in the form:

$$\begin{aligned}\dot{\bar{x}} &= A\bar{x} + B\bar{e} \\ y &= C\bar{x} + D\bar{e}\end{aligned}$$

where $\bar{x}$ is the state-vector, $\bar{e}$ is the input-vector, y is the output, and A, B, C and D are matrices of suitable dimension. Take the space vector as $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} i_L \\ v_C \end{bmatrix}$



(b) Given that input is $i_s = -20u(t)$, discretize the state-space model using forward-Euler approach (i.e. $\frac{dx}{dt} \sim \frac{x(n+1) - x(n)}{\Delta t}$) and the value of $\Delta t = 10 \mu s$. Obtain the value of state-vector: $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ for $n = 0, 1$, and 2 .

(a)

$$\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

$$\text{KCL at node 'a' gives } i_s = i_L + C \frac{dv_C}{dt} \Rightarrow \frac{dv_C}{dt} = -\frac{i_L}{C} + \frac{i_s}{C} \quad \dots (1)$$

$$\text{KVL in Loop-1 gives: } v_C - v_R = v_L \Rightarrow v_C - i_L \cdot R = L \frac{di_L}{dt}$$

2-marks for correct procedure.

$$\therefore \frac{di_L}{dt} = -\frac{R}{L} \cdot i_L + \frac{v_C}{L} \quad \dots (2)$$

$$\begin{bmatrix} \frac{di_L}{dt} \\ \frac{dv_C}{dt} \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & \frac{1}{L} \\ -\frac{1}{C} & 0 \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{C} \end{bmatrix} i_s$$

4-marks for correct entries for all the matrices (A,B,C,D).

Deduct 0.5 marks for each incorrect entry. Allow numerical entries in the answers.

$$\bar{v}_0 = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} i_s \quad \bar{x}(0) = \begin{bmatrix} i_L(0) \\ v_C(0) \end{bmatrix} = \begin{bmatrix} 10 \\ 200 \end{bmatrix}$$

(b) forward-Euler discretization gives the following.

$$i_L(n+1) = i_L(n) + \Delta t \left(-\frac{R}{L} \cdot i_L(n) + \frac{1}{L} v_C(n) \right)$$

$$v_C(n+1) = v_C(n) + \Delta t \left(-\frac{1}{C} i_L(n) + \frac{1}{C} i_s(n) \right)$$

$$\bar{x}(0) = \begin{bmatrix} i_L(0) \\ v_C(0) \end{bmatrix} = \begin{bmatrix} 10 \\ 200 \end{bmatrix}$$

2-marks for $\bar{x}(0)$

$$\bar{x}(1) = \begin{bmatrix} 10 \\ 170 \end{bmatrix}$$

2-marks for $\bar{x}(1)$

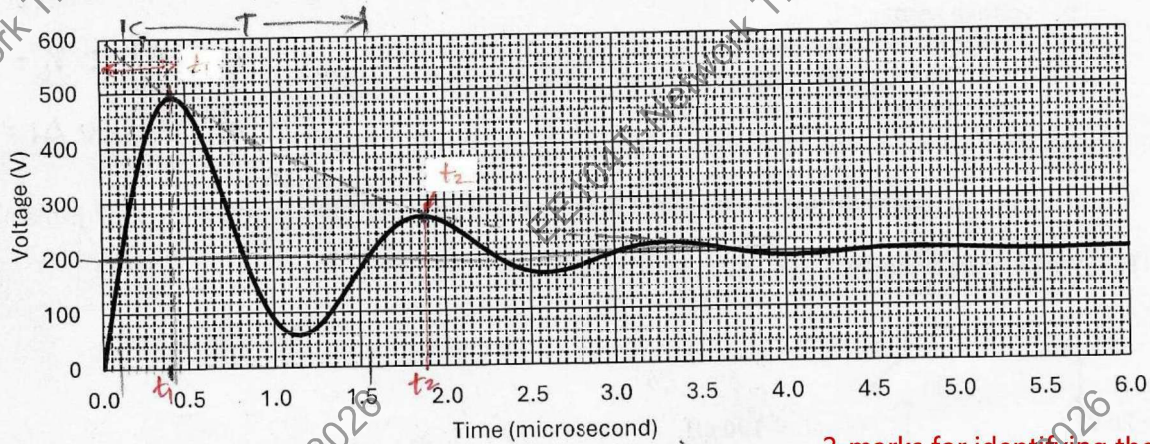
$$\bar{x}(2) = \begin{bmatrix} 9.7 \\ 140 \end{bmatrix}$$

2-marks for $\bar{x}(2)$

Deduct 1-mark for each incorrect entry in the vectors $\bar{x}(0), \bar{x}(1), \bar{x}(2)$.

Q8: Following plot shows the waveform of voltage across a capacitor (v_c) which is a part of a second-order network.

- Estimate the mathematical expression for this voltage (v_c). (8-marks)
[hint: 1. estimate the frequency of the oscillation 2. estimate the rate of decay of peak values and co-relate it to damping-coefficient.]
- Sketch the waveform of current through the capacitor. Mark the horizontal and vertical axes with approximate value of peaks, zero-crossings, steady-state value, etc. (Bonus problem - 4-marks).



(i) (a) $v_c(t \rightarrow \infty) = 200$ (Steady-state value)

2-marks for identifying the steady-state value

(b) $v_c = 200 + e^{-\alpha t} (A \cos(\omega_d t) + B \sin(\omega_d t))$

2-marks for identifying the under-damped response.

(c) from graph: $T = 1.45 \mu s \Rightarrow \omega_d = \frac{2\pi}{T} = \frac{2 \times 3.14}{1.45 \times 10^{-6}} = 4.33 \times 10^6 \text{ rad/s}$

the oscillating component of response decay with ' α '.

(d) $v_c(0) = 0 \Rightarrow 0 = 200 + 1(A + 0) \Rightarrow A = -200$

(e) We know that peaks decay exponentially at the rate ' α '

$$\begin{aligned} 300 &= K e^{-\alpha t_1} \\ 70 &= K e^{-\alpha t_2} \end{aligned} \Rightarrow \frac{300}{70} = e^{-\alpha(t_1 - t_2)} \Rightarrow 1.455 = e^{-\alpha(0.4 - 1.9)} \Rightarrow \alpha = 970,000 \text{ rad/s}$$

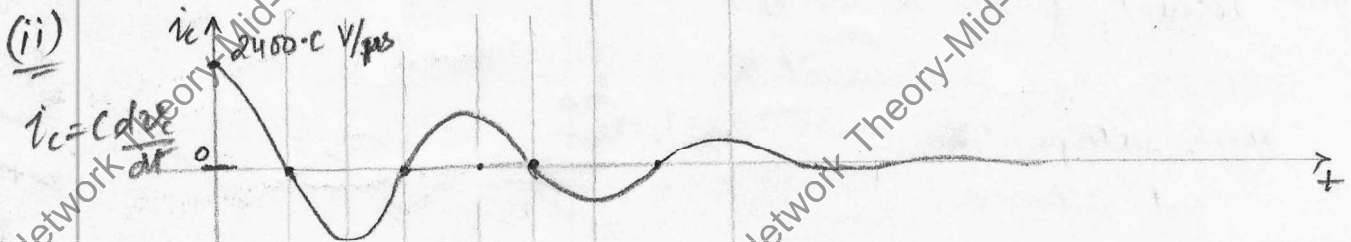
2-marks for rest of the procedure

(f) $v_c(0.4 \mu s) = 490 = 200 + e^{-\alpha t} (-200(-0.1605) + B(0.987))$

$\Rightarrow 290 = 0.6784(32.1 + 0.987B) \Rightarrow B = 400$

1-mark for correct value of angular frequency, and 1-mark if any of the other unknowns (alpha, A, B) are correct

$\therefore v_c = 200 + e^{-970000 t} (-200 \cos(4.33 \times 10^6 t) + 400 \sin(4.33 \times 10^6 t)) \text{ V.}$



Give 4-marks if the waveform looks like this.

Q9: The following figure (Fig. (a)) shows a circuit-representation of a power-electronic converter with ideal switches S1 and S2. The switches S1 and S2 are complementary to each other, that is, when S1 is ON then S2 is OFF, and when S1 is OFF then S2 is ON. The ON – OFF state of switch S1 with respect to time is given in Fig. (b). The value of capacitance is very-large, mean the voltage across it can be assumed to be constant in steady-state. In the steady-state, $v_i = 20$ V and $v_o = 10$ V. **(8-marks + 4-marks for Bonus problem)**

i. Sketch the waveform for the following signals with respect to time:

- voltage across S1, v_{S1}
- voltage across S2, v_{S2}
- voltage across inductor, v_L
- current through the inductor, i_L , given that $i_L(0) = 2$ A
- current through the resistor, i_R
- current through the capacitor, i_C **(Bonus problem)**

$$V_L = L \frac{di}{dt} : V_L = L \frac{\Delta i}{\Delta t}$$

$$\Rightarrow \Delta i = 2A$$

ii. Is there any contradiction to the initial assumption related to the capacitor voltage? Explain. **(Bonus problem)**

iii. Why are the switches S1 and S2 switching in complementary manner?

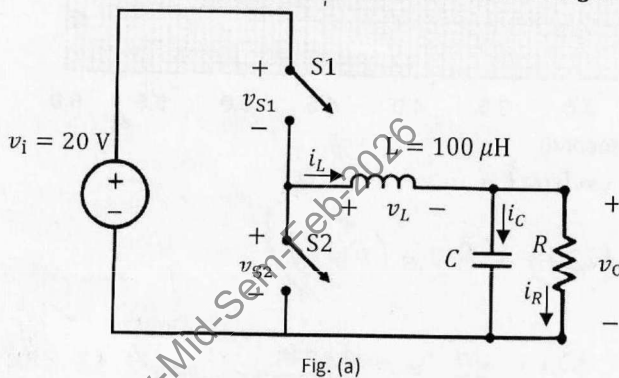
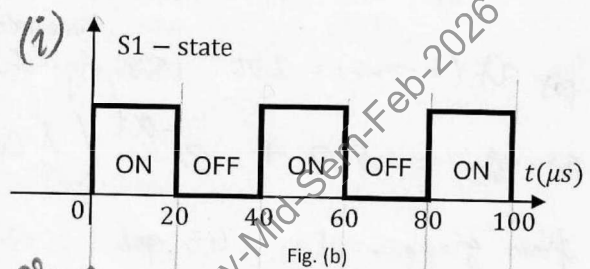


Fig. (a)

For any of the following, deduct 0.5 marks if the y-axes values are not mentioned.



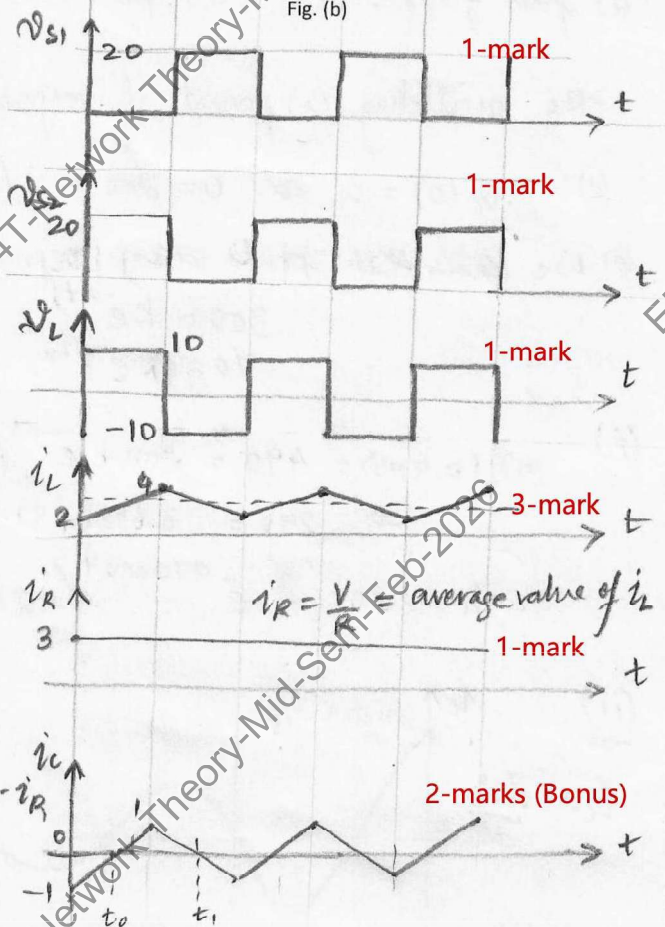
(ii) current through capacitor is rising & falling $\Rightarrow$ voltage across capacitor should also vary, since $i_C = C \frac{dv_C}{dt}$, however v_C is assumed to be constant! $\Rightarrow i_C = 0$

however: $\Delta v_C \approx \frac{1}{2} \left(\frac{di_C}{dt} \right) \cdot 20 \mu s / 100 \mu F$
typically $C \sim 100 \mu F \Rightarrow \Delta v = 0.1$ V
which is $\sim 1\%$ of 10V

2-marks for comment on the assumption. (Bonus)

(iii) Because, if they do not then S1 & S2 will short-circuit the supply voltage, v_i . Not allowed!

1-marks for correct response.



EE104T: Network Theory

Assignment-3

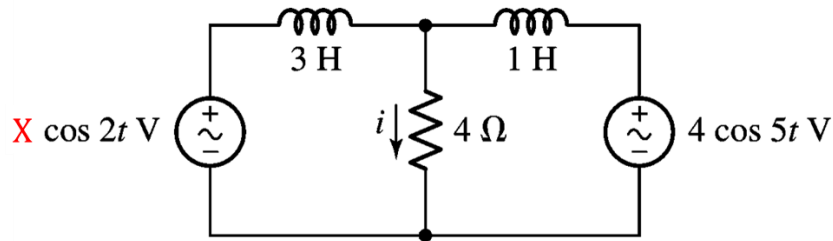
25th Mar' 2026

Total marks - 70

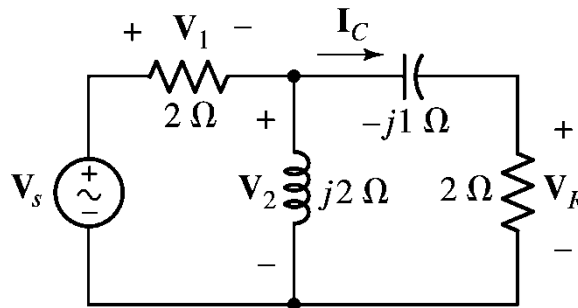
Submission deadline – 5th Apr' 2026**Important Note:**

- Many of the problems have value given by X . The value of X corresponding to your roll number is given on the last page (**Page-8**).
- Answers corresponding to the assigned value of X will only be considered.
- Provide your answers in the .txt file which is attached as a template. The answers should be rounded to the second decimal place.

1. Determine the peak-value (in mA) of current i during the steady-state in the following network. **(3-marks)**

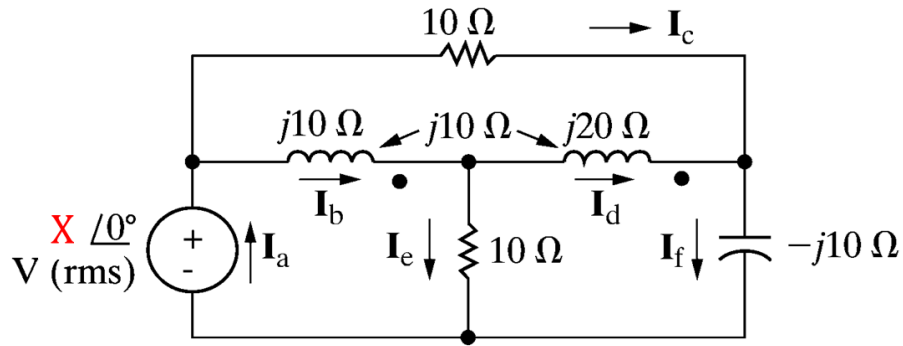


2. Given that $\mathbf{I}_C = \frac{X}{50} \angle 0^\circ$ A in following circuit. Assume that the quantities are represented by their RMS value. **(6-marks)** Draw a phasor diagram showing $\mathbf{V}_R$, $\mathbf{V}_2$, $\mathbf{V}_1$, and $\mathbf{V}_S$. **(ungraded)**
 (a) Give the ratios of the lengths of (a) $\mathbf{V}_S$ to $\mathbf{V}_1$ (b) $\mathbf{V}_1$ to $\mathbf{V}_2$; (c) $\mathbf{V}_S$ to $\mathbf{V}_R$.

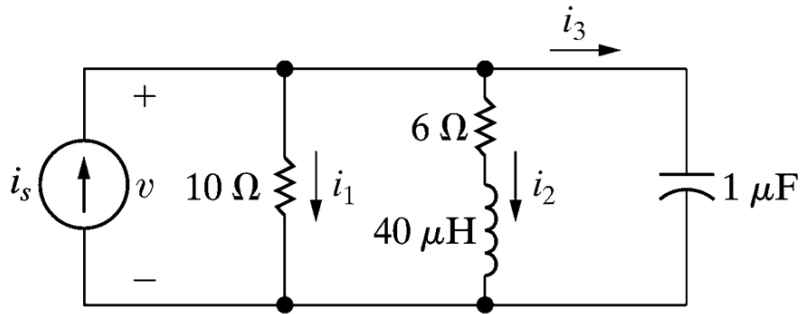


3. For the circuit shown below: **(12-marks)**

- (a) Find the six branch currents $I_a - I_f$ and give answer in the prescribed format.
 (b) Find the complex power absorbed by each branch of the circuit (**ungraded**)
 (c) Check your calculations by verifying the balance of complex power in the given network (**ungraded**)

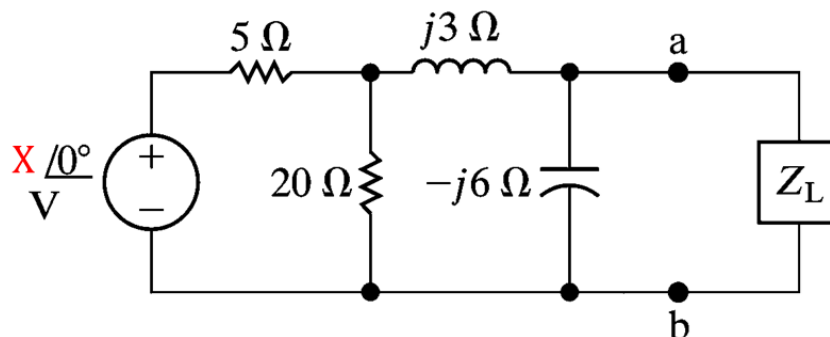


4. In the following circuit, $i_s = \sqrt{2} \cdot X \cdot \cos(200,000t)$ A. Determine the peak-values of v , i_1 , i_2 , and i_3 , in steady-state. Give their peak-values in mV and mA. **(8-marks)**

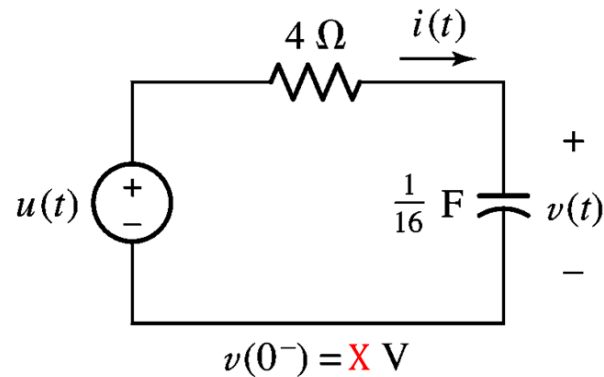


5. For the circuit shown below: **(4-mark)**

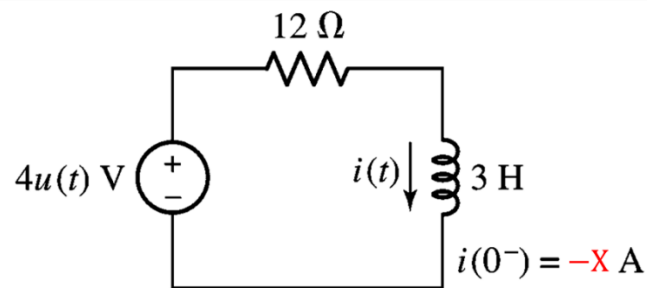
- (a) Determine the impedance that results in maximum average power transferred to Z_L
 (b) What is the maximum average power transferred to the load impedance determined in (a)?
 Assume that the quantities are represented by their RMS value.



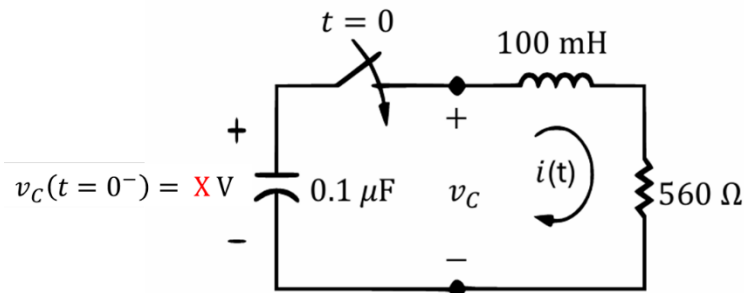
6. For the series-RC circuit shown below, determine $i(t)$ for $t > 0$ using s-domain technique. Give the maximum-value of the current i in mA as the response. **(3-marks)**



7. Determine $i(t)$ in the following circuit using s-domain technique. Give the maximum-value of the current i in mA as the response. **(3-mark)**

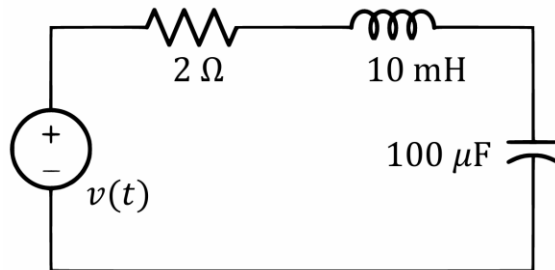


8. The switch in the given network is closed at $t = 0$. The voltage across the capacitor was X V at $t = 0$. Obtain the expression for $i(t)$ valid for $t \geq 0$ using s-domain approach. Give the maximum-value of the current i in mA as the response. **(3-marks)**



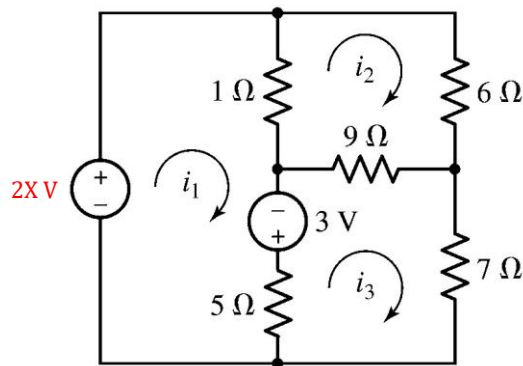
9. For the network shown below, $v(t) = \sqrt{2} \cdot X \cdot \sin(2\pi f t)$ V. **(6-marks)**

- Determine the value of f (frequency in Hz), that leads to maximum-power drawn from the source.
- Obtain the value of real-power supplied by the source for the determined frequency. (Hint: use Phasor Analysis) **(Ungraded)**



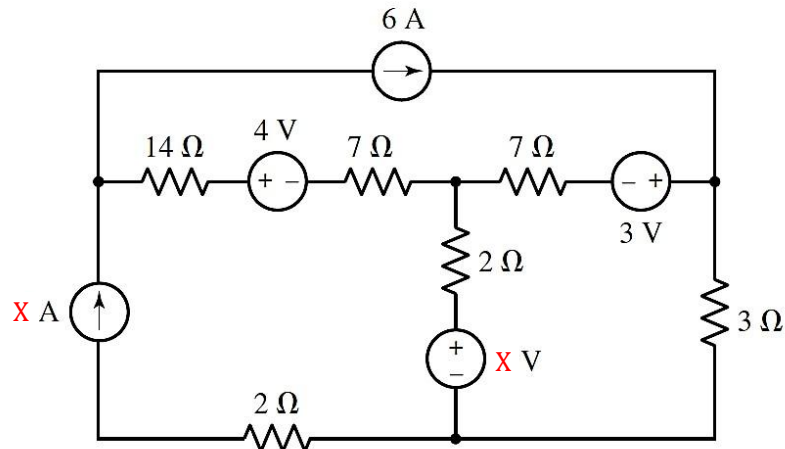
10. Draw the directed-graphs of the following network and obtain the following: **(8-marks)**

- Node-incidence matrix **A**, Give the values of determinant and trace of (AA^T) as your response.
- element impedance matrix Z_e , Give the values of determinant and trace of (Z_e) as your response
- element source-current matrix i_s , (Ungraded)
- element source-voltage matrix v_s , (Ungraded)
- B** and **Q** matrices by choosing a suitable tree. (Ungraded)

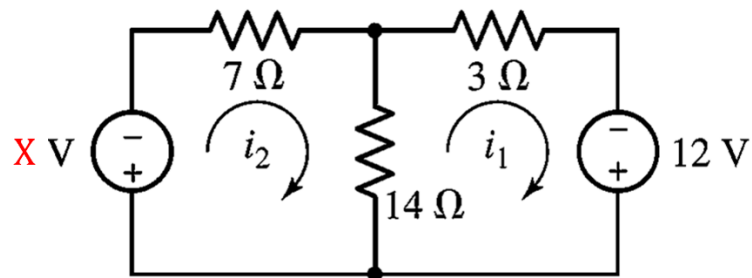


11. Draw the directed-graphs of the following network and obtain the following: **(8-marks)**

- Node-incidence matrix $\mathbf{A}$, Give the values of determinant and trace of $(\mathbf{A}\mathbf{A}^T)$ as your response.
- element impedance matrix $\mathbf{Z}_e$, Give the values of determinant and trace of $(\mathbf{Z}_e)$ as your response.
- element source-current matrix $\mathbf{i}_s$, (Ungraded)
- element source-voltage matrix $\mathbf{v}_s$, (Ungraded)
- $\mathbf{B}$ and $\mathbf{Q}$ matrices by choosing a suitable tree. (Ungraded)

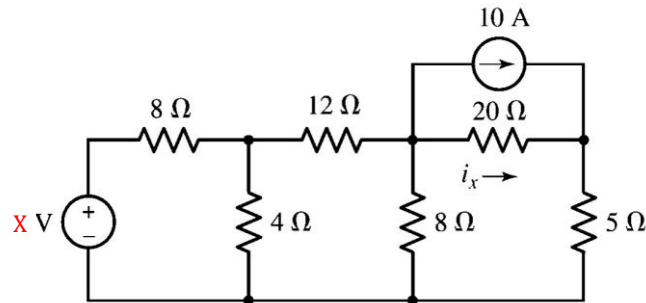


12. For the following circuit, obtain the currents i_1 and i_2 using graph-based approach. Give the peak-values of i_1 and i_2 in mA as your response. **(6-mark)**

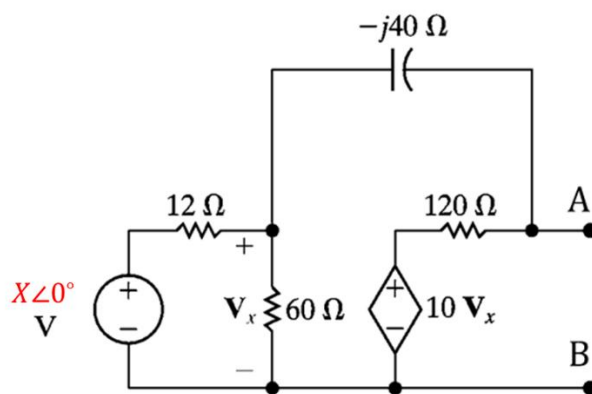


Practice-Problems (Ungraded)

1. Calculate the current i_x in the following circuit using graph-based approach. **(Ungraded)**

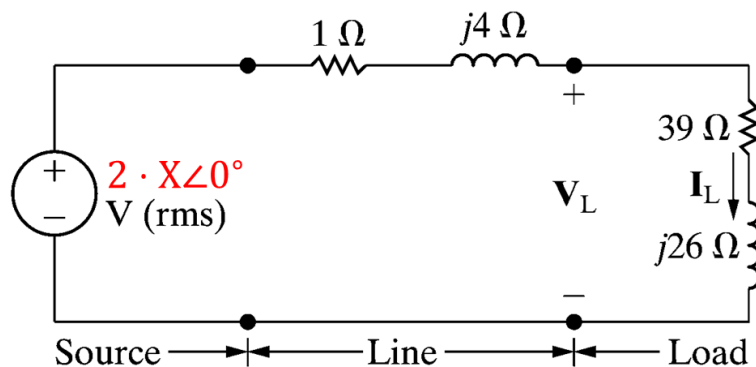


2. Find the Thevenin equivalent of the following network with respect to terminals A and B. Assume that the quantities are represented by their RMS value. **(Ungraded)**



3. In the circuit shown below, a load having an impedance of $39 + j26 \Omega$ is fed from a voltage source through a line having an impedance of $1 + j4 \Omega$. The effective, or rms, value of the source voltage is 250 V. **(Ungraded)**

- Calculate the load current I_L and voltage V_L .
- Calculate the complex power absorbed by the load.
- Calculate the complex power absorbed by the line.
- Calculate the complex power absorbed by the source.

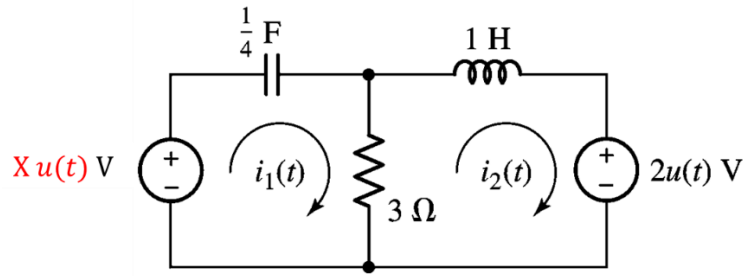


4. Obtain the time-domain expression for each of the following s-domain functions.

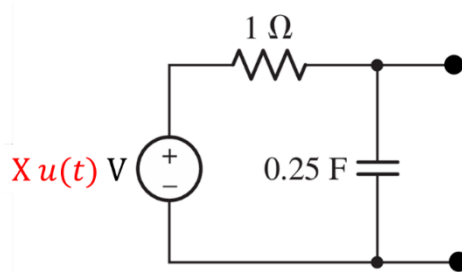
(Ungraded)

(a) $\frac{2}{s^2} + \frac{1}{s} + \frac{s+2}{\frac{s^2}{4} + 4s + 6}$ (b) $\frac{2}{(s+1)(s+1)}$ (c) $\frac{14}{(s+1)^2(s+4)(s+5)}$ (d) $7 - \frac{(s+\frac{1}{s})}{s^2+3s+1}$

5. Find the mesh currents i_1 and i_2 in the circuit shown below using s-domain method. Assume that no energy is stored in the circuit at $t = 0^-$. **(Ungraded)**



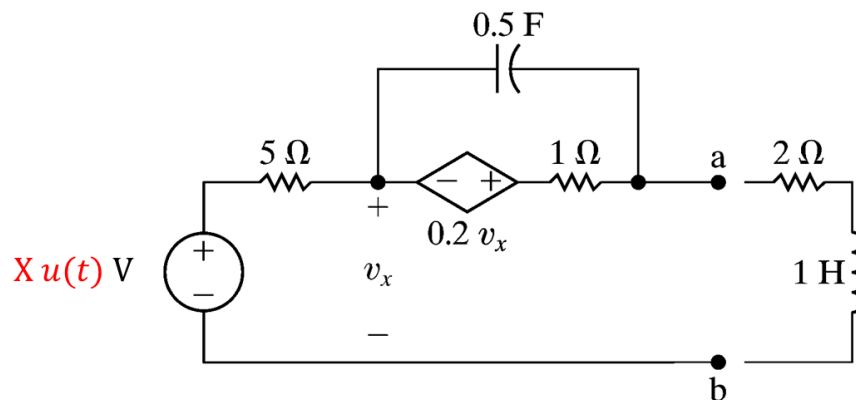
6. Using s-domain representation, find the Norton equivalent of the following network with respect to the open terminals. **(Ungraded)**



7. The initial charge on the capacitor in the following circuit is zero. **(Ungraded)**

(a) Obtain the s-domain Thevenin equivalent circuit with respect to terminals 'a' and 'b'.

(b) Obtain the s-domain expression for the current that the circuit delivers to a load consisting of a 1 H inductor in series with a 2 ohm resistor.



Roll No	X
EC25BT001	88
EC25BT002	98
EC25BT003	74
EC25BT004	109
EC25BT006	106
EC25BT007	27
EC25BT009	85
EC25BT010	32
EC25BT011	87
EC25BT012	119
EC25BT013	58
EC25BT014	53
EC25BT015	21
EC25BT016	70
EC25BT017	67
EC25BT018	95
EC25BT019	90
EC25BT020	108
EC25BT021	53
EC25BT022	31
EC25BT023	108
EC25BT024	105
EC25BT025	100
EC25BT026	81
EC25BT027	116
EC25BT028	91
EC25BT029	81
EC25BT030	45
EC25BT031	84
EC25BT033	101
EC25BT034	62
EC25BT035	51
EC25BT036	92
EC25BT037	73
EC25BT038	88
EC25BT039	115
EC25BT040	102
EC25BT041	91
EC25BT042	105
EC25BT043	31
EC25BT044	82

Roll No	X
EE25BT002	92
EE25BT003	60
EE25BT004	73
EE25BT005	53
EE25BT006	56
EE25BT007	67
EE25BT008	91
EE25BT009	24
EE25BT010	53
EE25BT011	42
EE25BT012	69
EE25BT013	69
EE25BT014	46
EE25BT015	41
EE25BT018	36
EE25BT020	82
EE25BT021	108
EE25BT022	24
EE25BT023	113
EE25BT024	110
EE25BT025	63
EE25BT026	99
EE25BT028	64
EE25BT029	54

Quiz-2 – EE104T: Network Theory

39
21

Q1: Show the poles and zeroes of the following transfer functions on the s-plane. (5 marks)

(a) $V(s) = \frac{2}{s} + \frac{s+3}{s^2+4s+3}$

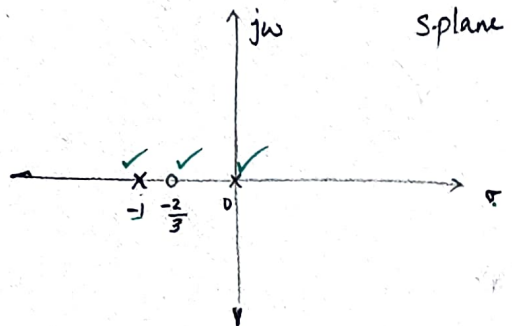
(b) $I(s) = \frac{s+2}{s^2+4s+68}$

(a) $V(s) = \frac{2}{s} + \frac{s+3}{(s+3)(s+1)} = \frac{2}{s} + \frac{1}{s+1} = \frac{3s+2}{s(s+1)}$

Poles: $s=0$ & $s=-1$

Zeros: $s = -\frac{2}{3}$

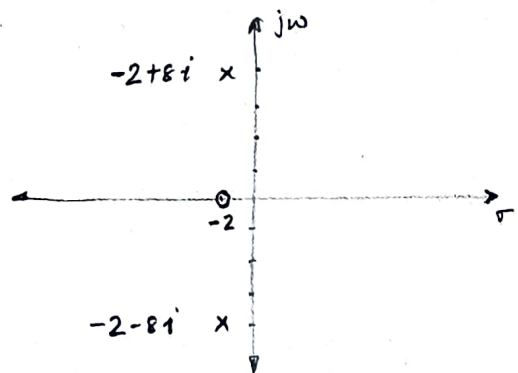
1-mark each for correct poles and zeros



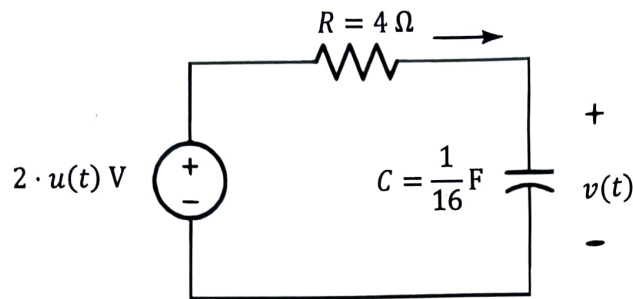
(b) $I(s) = \frac{s+2}{s^2+4s+68} = \frac{s+2}{(s+(2+8i))(s+(2-8i))}$

poles: $s = -2-8i$, $s = -2+8i$

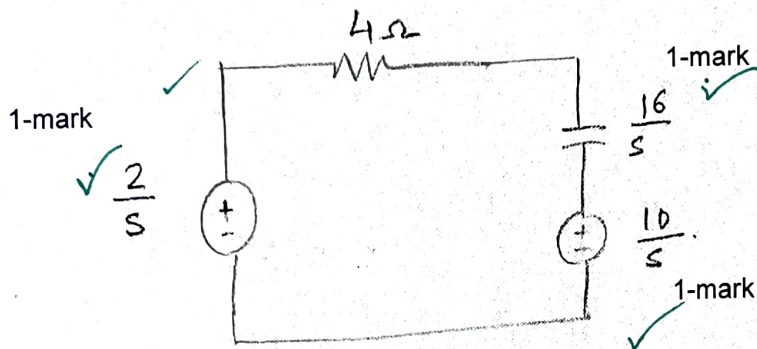
1-mark each for correct poles



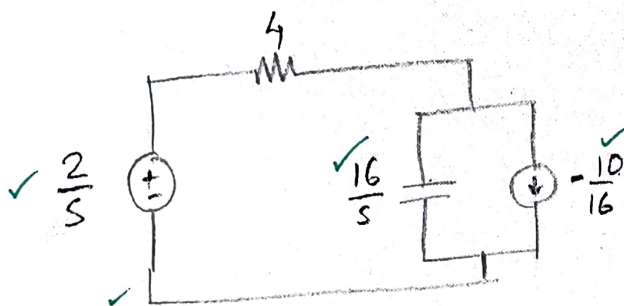
Q4: Draw the s-domain equivalent model of the following network that can be used for obtaining the response $i(t)$ in this network. The initial voltage across the capacitor is $v(0^-) = 10 \text{ V}$ (3-marks)



Any of the following two models are correct.

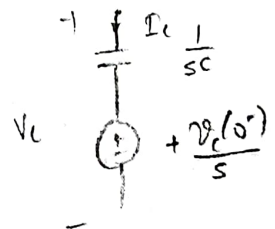
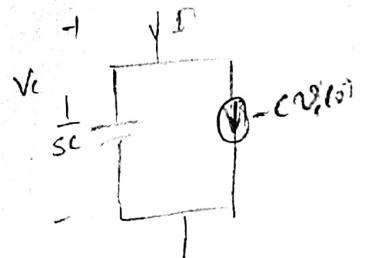


OR

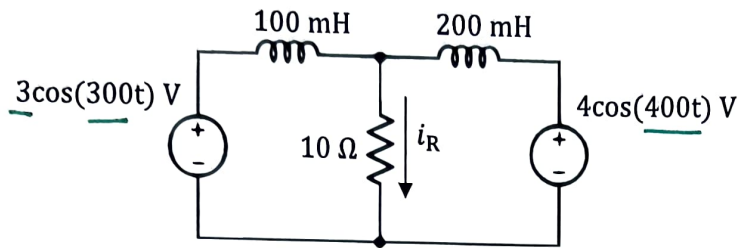


$$i_c = C \frac{dv_c}{dt}$$

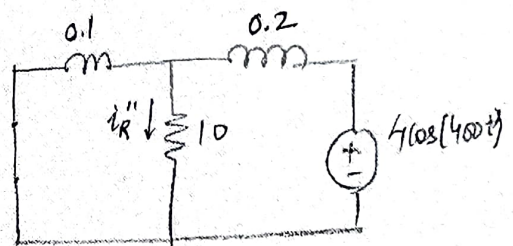
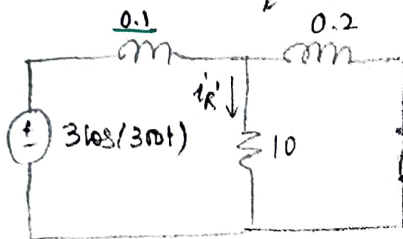
$$I = s C V_c - C v_c(0^-)$$



Q2: In the given circuit, determine the expression for current i_R during the steady-state. Mention the peak-value of the current i_R . (5 marks)

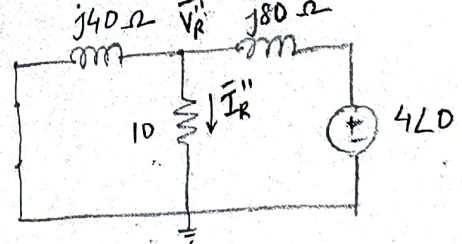
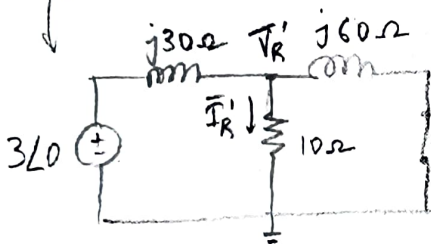


Superposition



Phasor

Phasor



$$\bar{I}_R' = \frac{\bar{V}_R'}{10} = \frac{(10 \parallel j60) \times 3\angle 0}{j30 + (10 \parallel j60)} \cdot \frac{1}{10}$$

$$\bar{I}_R'' = \frac{\bar{V}_R''}{10} = \frac{(10 \parallel j40) \cdot 4\angle 0}{(10 \parallel j40) + j80} \cdot \frac{1}{10}$$

$$= -0.04 - 0.08j$$

$$= 0.0894 \angle -63.43^\circ$$

$$= 0.0164 - 0.0438j$$

$$\bar{I}_R'' = 0.0468 \angle -69.44^\circ$$

If the student has directly written $\bar{I}_R = \bar{I}_R' + \bar{I}_R''$, then directly give 0 for the complete problem, even if the final answer is correct.

$$i_R' = 0.0894 \cos(300t - 63.43^\circ)$$

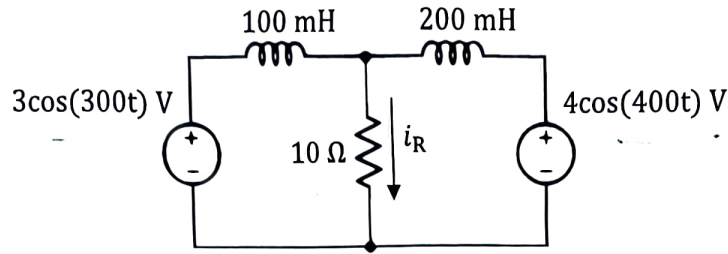
$$i_R'' = 0.0468 \cos(400t - 69.44^\circ)$$

Finally, using superposition $i_R = i_R' + i_R'' = 0.0894 \cos(300t - 63.43^\circ) + 0.0468 \cos(400t - 69.44^\circ)$

No partial marks, give 3-marks for correct response. Deduct 1-mark for each incorrect entry (amplitude/phase)

Q3: Int branch branch

Q2: In the given circuit, determine the expression for current i_R during the steady-state. Mention the peak-value of the current i_R . (5 marks)



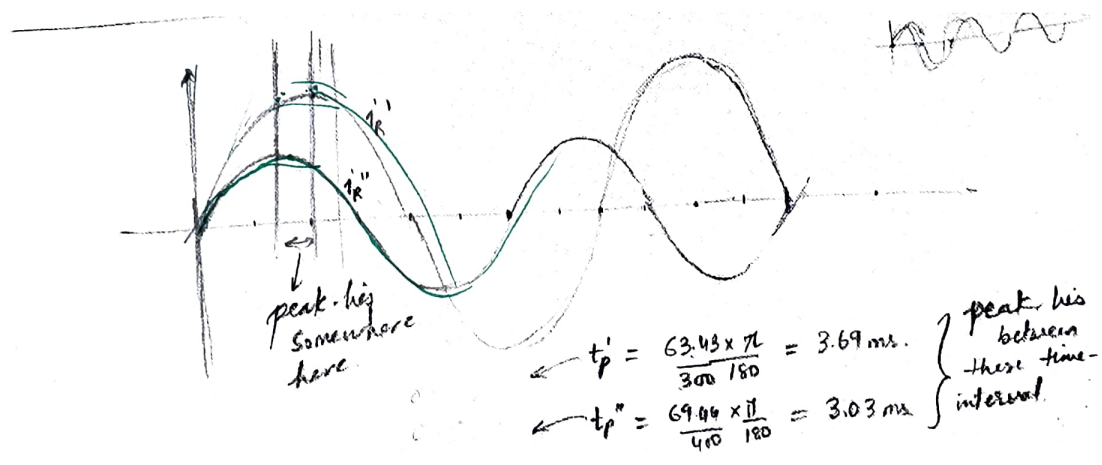
Finally, using superposition $i_R = i_R' + i_R'' = 0.0894 \cos(300t - 63.43^\circ) + 0.0468 \cos(400t - 69.44^\circ)$

peak-value of i_R

$\frac{di_R}{dt} = 0 \Rightarrow 0 = -0.0894 \cdot \sin(300t - 63.43^\circ)(300) - 0.0468 \cdot 400 \cdot \sin(400t - 69.44^\circ)$
 $\rightarrow$ Requires solution of Non-linear Equation (1)

The problem can be approached graphically with some simplification.

Since the phase angles of superposed waveforms are close, the waveform will be as follows:



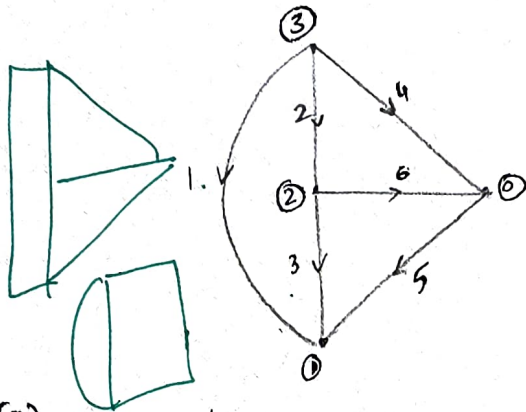
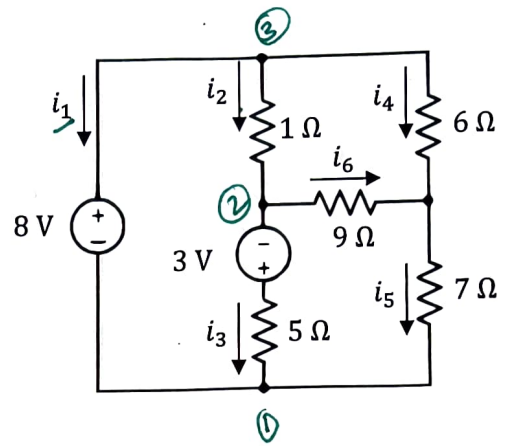
The peak-value will lie somewhere between the peaks of superposed waveforms.

Give 1-marks for approach, and 1-marks for correct final value ~0.134 - 0.135 A.

0.136 A

Q3: In the shown circuit, the direction of currents in each branch is denoted. The subscript of current represents the branch-number. Draw the directed-graph of this network using the branch-numbers mentioned in the figure, and then obtain the following: (7-marks)

- Node-incidence matrix A
- element impedance matrix Z_e
- element source-voltage matrix v_s
- The B and Q matrices by choosing a tree that include the branches containing $1\ \Omega$, $5\ \Omega$, and $9\ \Omega$ resistors.



(a)

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \textcircled{1} & -1 & 0 & -1 & 0 & -1 \\ \textcircled{2} & 0 & -1 & 1 & 0 & 0 \\ \textcircled{3} & 1 & 1 & 0 & 1 & 0 \\ \textcircled{4} & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1-mark for correct A matrix. Deduct 0.25 marks for each incorrect entry. The rows may be interchanged based on the node-numbering.

(c)

$$v_s = \begin{bmatrix} 1 & 8 \\ 2 & 0 \\ 3 & -3 \\ 4 & 0 \\ 5 & 0 \\ 6 & 0 \end{bmatrix}$$

only in order

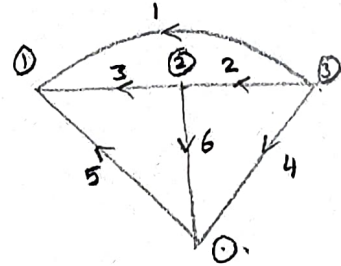
(c)

1-mark for correct v_s matrix. Deduct 0.25 marks for each incorrect entry

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \textcircled{1} & 1 & -1 & -1 & 0 & 0 \\ \textcircled{2} & 0 & -1 & 0 & 1 & 0 \\ \textcircled{3} & 0 & 0 & -1 & 0 & 1 \end{bmatrix}$$

2-mark for correct B matrix. Deduct 0.25 marks for each incorrect entry. The rows may be interchanged based on the fundamental-loop numbering.

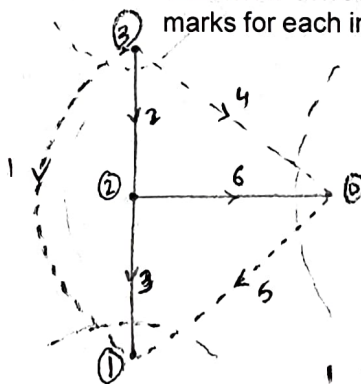
OR



(b)

$$Z_e = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \textcircled{1} & 0 & & & & \\ \textcircled{2} & & 1 & & & \\ \textcircled{3} & & & 5 & & \\ \textcircled{4} & & & & 6 & \\ \textcircled{5} & & & & & 7 \\ \textcircled{6} & & & & & & 9 \end{bmatrix}$$

1-mark for correct Z matrix. Deduct 0.25 marks for each incorrect entry



(c)

$$Q = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ \textcircled{CS2} & -1 & 1 & 0 & -1 & 0 \\ \textcircled{CS3} & 1 & 0 & 1 & 0 & 1 \\ \textcircled{CS6} & 0 & 0 & 0 & 1 & -1 \end{bmatrix}$$

2-mark for correct Q matrix. Deduct 0.25 marks for each incorrect entry. The rows may be interchanged based on the cutset-numbering.

EE104T: Network Theory

Assignment – 4

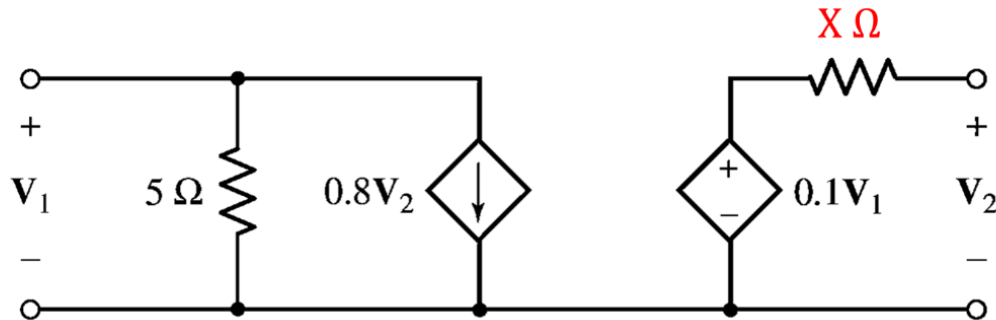
21st Apr 2026

Total marks - 40

Submission deadline – 26th Apr 2026**Important Note:**

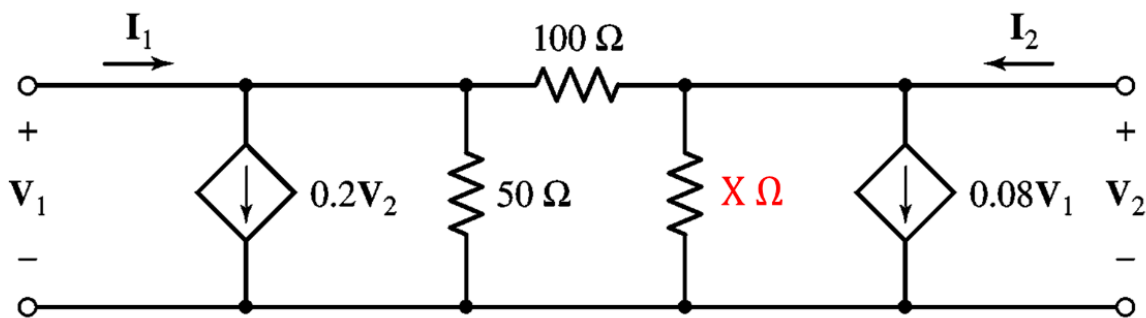
- Many of the problems have value given by X . The value of X corresponding to your roll number is given on the last page (Page-4).
- Answers corresponding to the assigned value of X will only be considered.
- Provide your answers in the .txt file which is attached as a template. Enter your answers in SI-units with three significant figures.

- Determine the impedance parameters for the following two-port network. (8-mark)



- Obtain the impedance and admittance parameters for the two-port network shown below. Enter the determinant of impedance parameter matrix and determinant of admittance parameter matrix as your responses.

(8-mark)



3. Two sets of measurements are made on a two-port resistive circuit. The first set is made with port 2 open, and the second set is made with port 2 short-circuited.

The results are as follows:

Port 2 Open	Port 2 Short-circuited
$V_1 = 10 \text{ mV}$	$V_1 = \text{X mV}$
$I_1 = 10 \text{ } \mu\text{A}$	$I_1 = 20 \text{ } \mu\text{A}$
$V_2 = -40 \text{ V}$	$I_2 = 1 \text{ mA}$

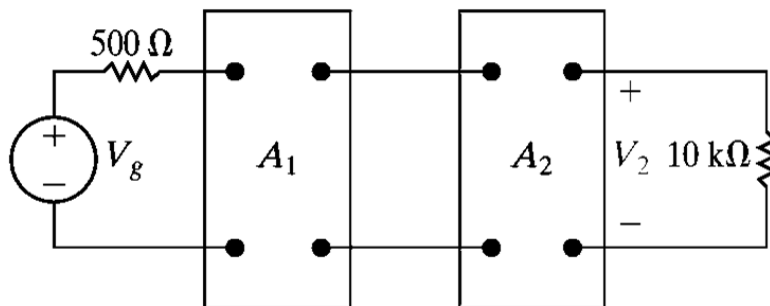
Obtain the **h**-parameters of the circuit. **(8-mark)**

4. The following measurements are made on a two-port circuit operating in the sinusoidal steady state.
 With port 2 open, a voltage equal to $\text{X} \cos 4000t \text{ V}$ is applied to port 1. The current into port 1 is $25 \cos(4000t - 45^\circ) \text{ A}$, and the port 2 voltage is $100 \cos(4000t + 15^\circ) \text{ V}$.
 With port 2 short-circuited, a voltage equal to $30 \cos(4000t) \text{ V}$ is applied to port 1. The current into port 1 is $1.5 \cos(4000t + 30^\circ)$ and the current into port 2 is $0.25 \cos(4000t + 150^\circ) \text{ A}$.

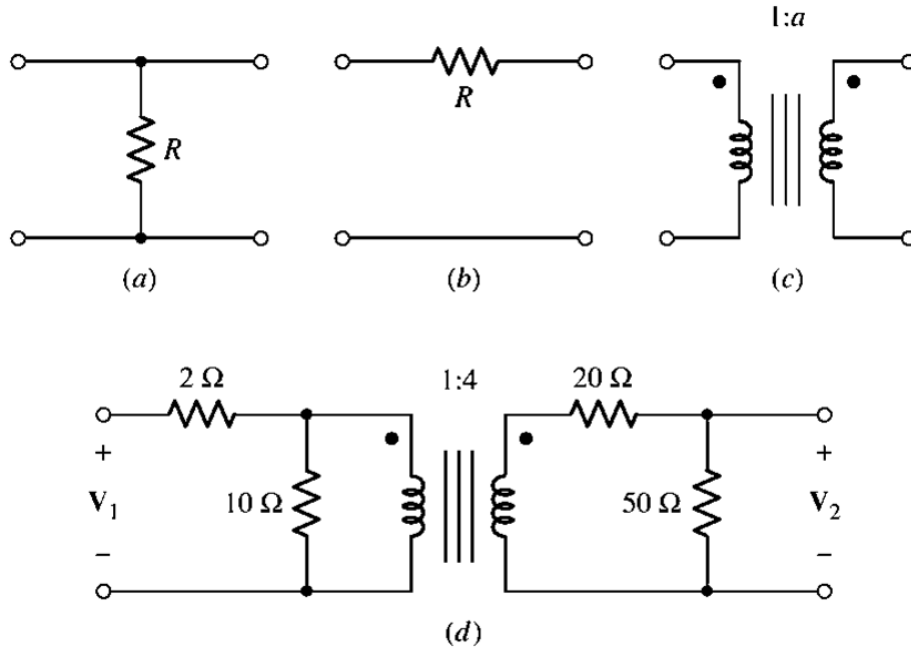
Find the **ABCD**-parameters that can describe the sinusoidal steady-state behaviour of the circuit. **(8-mark)**

5. Two identical amplifiers are connected in cascade, as shown in following figure. Each amplifier is described in terms of its h parameters. The values are $h_{11} = 100 \text{ } \Omega$, $h_{12} = 0.0015$, $h_{21} = \text{X}$, and $h_{22} = 100 \text{ } \mu\text{S}$.

Find the voltage gain V_2/V_g for the following network. **(8-mark)**



6. If $\mathbf{h}$ -parameters for some particular two-port network are given as $\mathbf{h} = \begin{bmatrix} 2 \text{ k}\Omega & -3 \\ 5 & 0.01 \text{ S} \end{bmatrix}$, calculate (a) $\mathbf{z}$ -parameters and (b) $\mathbf{y}$ -parameters. **(Ungraded)**
7. Find $\mathbf{t}$ -parameters of networks shown in (a), (b), and (c), in the following figure. Using the rules of interconnecting two-parts in cascade, find the $\mathbf{t}$ -parameters for the network shown in (d). **(Ungraded)**



Roll No	X
EC25BT001	88
EC25BT002	98
EC25BT003	74
EC25BT004	109
EC25BT006	106
EC25BT007	27
EC25BT009	85
EC25BT010	32
EC25BT011	87
EC25BT012	119
EC25BT013	58
EC25BT014	53
EC25BT015	21
EC25BT016	70
EC25BT017	67
EC25BT018	95
EC25BT019	90
EC25BT020	108
EC25BT021	53
EC25BT022	31
EC25BT023	108
EC25BT024	105
EC25BT025	100
EC25BT026	81
EC25BT027	116
EC25BT028	91
EC25BT029	81
EC25BT030	45
EC25BT031	84
EC25BT033	101
EC25BT034	62
EC25BT035	51
EC25BT036	92
EC25BT037	73
EC25BT038	88
EC25BT039	115
EC25BT040	102
EC25BT041	91
EC25BT042	105
EC25BT043	31
EC25BT044	82

Roll No	X
EE25BT002	92
EE25BT003	60
EE25BT004	73
EE25BT005	53
EE25BT006	56
EE25BT007	67
EE25BT008	91
EE25BT009	24
EE25BT010	53
EE25BT011	42
EE25BT012	69
EE25BT013	69
EE25BT014	46
EE25BT015	41
EE25BT018	36
EE25BT020	82
EE25BT021	108
EE25BT022	24
EE25BT023	113
EE25BT024	110
EE25BT025	63
EE25BT026	99
EE25BT028	64
EE25BT029	54

EE104T: Network Theory

(Instructor: Amarkumar Kushiwaha)

Full Name (All caps): _____, Roll No: _____

27th April 2026, 9:30 AM – 12:30 PM

Total marks: 70 Duration: 180 minutes

Instructions:

- There are total **11** questions and key-formulae on **4-printed sides**.
- Begin a **new-question from new-page**, clearly mentioning the question number.
- **No clarification to be sought during the exam**. If any information is missing, make **reasonable assumptions** and state the same with justification.
- Except for scientific calculators, **NO Electronics gadgets** are allowed.
- **Mention the unit of all the quantities in the intermediate-steps and the final-answers.**

Q1: Fill in the blanks by selecting the correct option. (15-marks)

1. In an ideal transformer, the ratio of primary to secondary voltage is equal to the _____.
(a. resistance ratio, ~~b. turns ratio~~, c. power ratio, d. frequency ratio)
2. In graph theory, a tree is a sub-graph that connects all nodes but contains no _____.
(a. branches, b. links, ~~c. loops~~, d. twigs)
3. A Gyrator is a non-reciprocal device that can transform a capacitor into a/an _____.
(a. resistor, ~~b. inductor~~, c. voltage source, d. transformer)
4. The Transfer Function $H(s)$ of a system is defined as the ratio of the Laplace transform of the output to the input, assuming _____ initial conditions.
(a. maximum, b. unit, ~~c. zero~~, d. infinite)
5. In a two-port network, the condition for reciprocity in terms of Z-parameters is _____.
(a. $Z_{11} = Z_{12}$, ~~b. $Z_{12} = Z_{21}$~~ , c. $Z_{11} = Z_{22}$, d. $Z_{22} = Z_{21}$)
6. The incidence matrix of a graph represents the relationship between _____ and branches
(a. loops, ~~b. nodes~~, c. chords, d. meshes)
7. According to Tellegen's Theorem, the sum of instantaneous power in all branches of any network is _____.
(a. maximum, b. infinite, ~~c. zero~~, d. dependent on the source)
8. In t – parameters, the parameter t_{21} has the unit of _____.
(~~a. mhos~~, b. dimensionless, c. ohms, d. watts)
9. In graph theory, the branches of a graph that are not part of a tree are called _____.
(a. twigs, ~~b. links~~, c. nodes, d. paths)
10. For a Capacitor, the s-domain equivalent circuit consists of an impedance $\frac{1}{sC}$ in series with a voltage source of value _____.
(a. CV_0 , ~~b. $\frac{V_0}{s}$~~ , c. sV_0 , d. $\frac{I_0}{C}$)
11. A 10:1 ideal transformer has a secondary load of $5\ \Omega$. The impedance referred to the primary is _____.
(a. $0.5\ \Omega$, b. $50\ \Omega$, ~~c. $500\ \Omega$~~ , d. $5\ \Omega$)
12. If a graph has 6 nodes and 10 branches, the number of links in any tree is _____.
(a. 4, ~~b. 5~~, c. 6, d. 10)
13. In a two-port network, the y – parameters are also known as _____ parameters.
(a. open-circuit impedance, ~~b. short-circuit admittance~~, c. hybrid, d. transmission)
14. A gyrator with gyration resistance $R = 100\ \Omega$ is terminated by a $10\ \mu\text{F}$ capacitor. The equivalent input inductance is _____.
(a. 10mH , ~~b. 100mH~~ , c. 1H , d. 0.1mH)
15. The maximum power transfer theorem states that for maximum power, the load impedance should be the _____ of the source impedance.
(a. equal, b. negative, ~~c. complex conjugate~~, d. reciprocal)

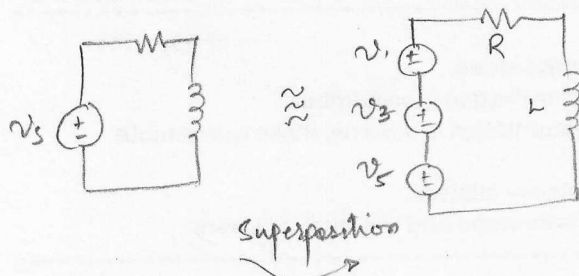
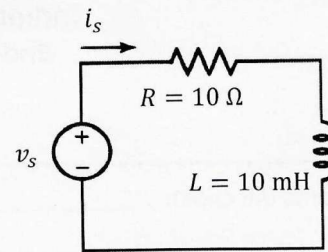
1-mark for each correct response

Q2. In the given network, a triangular voltage v_s is applied across a series combination of a resistor and an inductor. This wave can be approximated using a sum of sine-waves (fourier-series), as follows:

$$v_s \approx \frac{8}{\pi^2 1^2} \sin(314t) - \frac{8}{\pi^2 3^2} \sin(3 \cdot 314t) + \frac{8}{\pi^2 5^2} \sin(5 \cdot 314t) \text{ V.}$$

Obtain the expression for i_s during steady-state. **(6-marks)**

[hint: Phasor analysis]



$$v_1 = \frac{8}{\pi^2} \sin(314t) \text{ V}$$

$$v_3 = -\frac{8}{\pi^2 \cdot 9} \sin(3 \cdot 314t) \text{ V}$$

$$v_5 = \frac{8}{\pi^2 \cdot 25} \sin(5 \cdot 314t) \text{ V}$$

1-mark for identifying superposition approach

$$\bar{V}_1 = \frac{8}{\pi^2} \angle 0^\circ, \omega_1 = 314 \text{ rad/s} \Rightarrow \bar{I}_1 = \frac{\bar{V}_1}{Z_1} = \frac{8/\pi^2}{10 + j\omega L} = \frac{8/\pi^2}{10 + j3.14} = 0.0774 \angle -17.43^\circ \text{ A}$$

$$\therefore i_1 = 0.0774 \sin(314t - 17.43^\circ) \text{ A}$$

2-mark for correct procedure

$$\bar{V}_3 = \frac{+8}{\pi^2 \cdot 9} \sin(3 \times 314t + 180^\circ), \omega_3 = 3 \times 314 \Rightarrow \bar{I}_3 = \frac{\bar{V}_3}{Z_3} = \frac{8/\pi^2 \cdot 9}{10 + j\omega L} = 0.00656 \angle -43.3^\circ \text{ A}$$

$$\therefore i_3 = 0.00656 \sin(3 \cdot 314t + 136.7^\circ) \text{ A}$$

$$\text{or } i_3 = 0.00656 \sin(3 \cdot 314t - 223.3^\circ) \text{ A}$$

$$\bar{V}_5 = \frac{8}{\pi^2 \cdot 25} \angle 0^\circ, \omega_5 = 5 \times 314 \Rightarrow \bar{I}_5 = \frac{\bar{V}_5}{Z_5} = \frac{8/\pi^2 \cdot 25}{10 + j\omega L} = 0.00174 \angle -57.5^\circ$$

$$\therefore i_5 = 0.00174 \sin(5 \cdot 314t - 57.5^\circ) \text{ A}$$

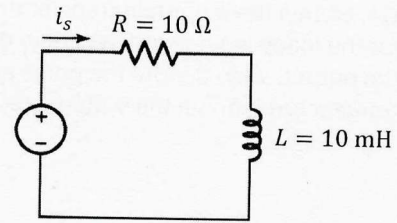
Finally

$$i_s = i_1 + i_3 + i_5 = 0.0774 \sin(314t - 17.43^\circ) + 0.00656 \sin(3 \cdot 314t + 136.7^\circ) + 0.00174 \sin(5 \cdot 314t - 57.5^\circ)$$

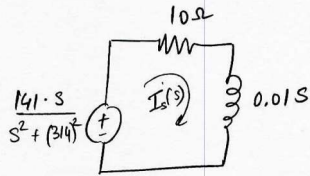
3-marks for correct final expression. Deduct 0.5 marks for each incorrect amplitude or phase angle

Q3. Using s-domain approach, obtain the expression for $i_s(t)$ for the circuit shown in this figure. (6-mark)

$$v_s(t) = 141 \cos(314t) u(t)$$



s-domain



$$\begin{aligned} I(s) &= \left(\frac{141 \cdot s}{s^2 + (314)^2} \right) / (10 + 0.01s) \\ &= 141 \left(\frac{100s}{(s^2 + 314^2)(s + 1000)} \right) \\ &= 141 \left(\frac{As + B}{s^2 + 314^2} + \frac{C}{s + 1000} \right) \end{aligned}$$

2-marks for correct approach

$$\begin{aligned} As^2 + Bs + 1000As + 1000B + 1000Cs + 314^2C &= 100s \\ \rightarrow A + C &= 0, \rightarrow C = -A \\ B + 1000A &= 100 \\ 1000B + 314^2C &= 0 \\ 1000B - 98596A &= 0 \rightarrow B = 98.596A \\ 98.596A + 1000A &= 100 \rightarrow A = 0.091 \\ B &= 8.975 \\ C &= -0.091 \end{aligned}$$

$$\therefore I(s) = 141 \left(\frac{0.091s}{s^2 + 314^2} + \frac{8.975}{s^2 + 314^2} - \frac{0.091}{s + 1000} \right)$$

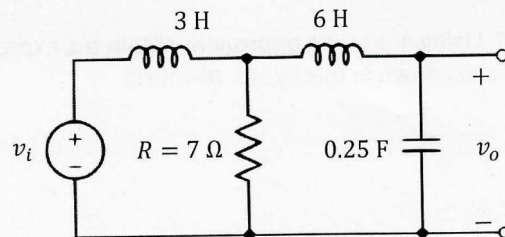
2-marks for obtaining this expression in s-domain

$$\therefore i(t) = \mathcal{L}^{-1}\{I(s)\} = (12.831 \cos(314t) + 4.03 \sin(314t) - 12.831 e^{-1000t}) u(t)$$

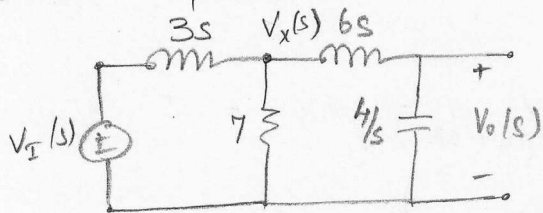
2-marks for correct time-domain expression. Deduct 0.5 marks for each incorrect entry in terms of coefficient, frequency, or exponent.

Deduct 1-mark if $u(t)$ is not mentioned.

Q4. Obtain the s-domain transfer function (output/input) for the following network where v_i is the input and v_o is the output. Also denote the poles and zeroes of this transfer function on the s-plane. **(6-mark)**



s-domain representation



1-mark for s-domain equivalent circuit

$$V_O(s) = \frac{4/s}{4/s + 6s} \cdot V_X \quad \& \quad V_X(s) = \frac{7 \parallel (6s + 4/s)}{7 \parallel (6s + 4/s) + 3s} \cdot V_I(s)$$

$$V_O(s) = \frac{4}{4 + 6s^2} \cdot V_X \quad \& \quad V_X(s) = \frac{7 \cdot (6s + 4/s)}{(7 + 6s + 4/s) + 3s} \cdot V_I(s)$$

2-marks for correct procedure

$$= \frac{7 \cdot (6s^2 + 4) / s}{7 \cdot (6s^2 + 4) + 3s \cdot s \cdot (7 + 6s + 4/s)} \cdot V_I(s)$$

$$V_X(s) = \frac{7 \cdot (6s^2 + 4)}{7 \cdot (6s^2 + 4) + 3s(7s + 6s^2 + 4)} \cdot V_I(s)$$

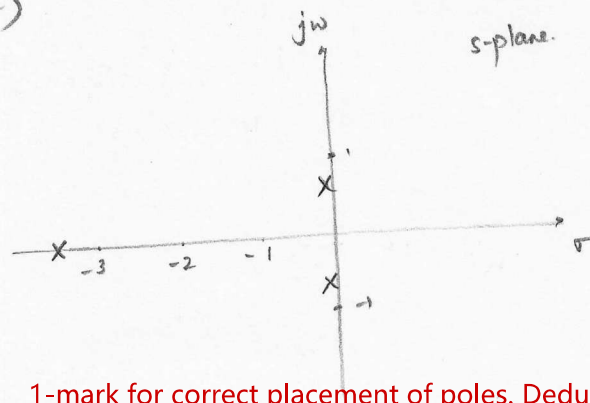
$$= \frac{7(6s^2 + 4)}{42s^2 + 28 + 21s^2 + 18s^3 + 12s} V_I(s)$$

$$\therefore V_O(s) = \frac{4}{(4 + 6s^2)} \cdot \frac{7(6s^2 + 4)}{(18s^3 + 63s^2 + 12s + 28)} \cdot V_I(s)$$

$$\frac{V_O(s)}{V_I(s)} = \frac{28}{18s^3 + 63s^2 + 12s + 28}$$

2-marks for correct final expression

poles: $s_1 = -3.438$
 $s_2 = -0.031 + j0.672$
 $s_3 = -0.031 - j0.672$



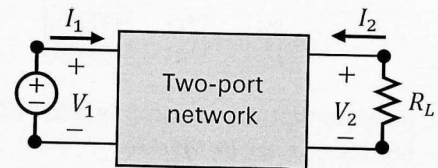
1-mark for correct placement of poles. Deduct 0.5 marks for each incorrect representation.

Q5: The two-port network shown here is represented as follows:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 5 & -2.5 \\ -2.5 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

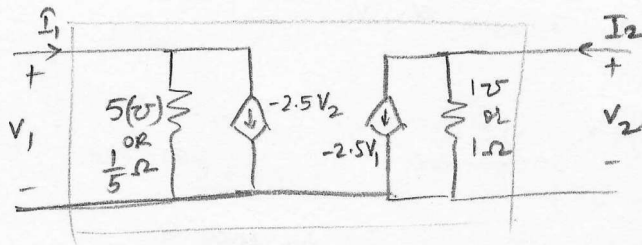
(a) Draw a network using electrical-circuit elements that represents this two-port network. (2-marks)

(b) Obtain the Thevenin equivalent of the complete network seen by resistor R_L , i.e. from port-2 excluding R_L . (4-marks)



(a) $I_1 = 5 \cdot V_1 - 2.5 V_2$

$I_2 = -2.5 V_1 + 1 \cdot V_2$



2-marks for correct representation.

deduct 0.5 marks for each incorrect notation/value.

(b) To obtain the thevenin equivalent, we disconnect ' R_L ' $\Rightarrow I_2 = 0$
 V_{th} : then obtain $V_2 \rightarrow V_{oc} \rightarrow V_{th}$

$$\begin{cases} I_1 = 5V_1 - 2.5V_2 \\ 0 = -2.5V_1 + V_2 \end{cases} \Rightarrow \begin{cases} I_1 = 5(\frac{V_2}{2.5}) - 2.5V_2 \\ \Rightarrow I_1 = -0.5V_2 \Rightarrow \underline{V_2 = 2I_1} \end{cases}$$

1-mark for correct procedure.

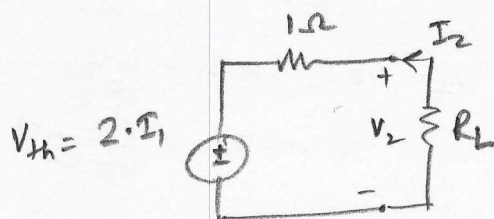
$\therefore V_{th} = 2I_1$

1-mark for correct expression for V_{th}

R_{th} : $R_{th} = \left. \frac{V_2}{I_2} \right|_{V_1=0} \Rightarrow I_2 = -2.5(0) + 1 \cdot V_2 = V_2$
 $\Rightarrow \frac{V_2}{I_2} = 1 \Omega = R_{th}$

2-marks for correct expression of R_{th}

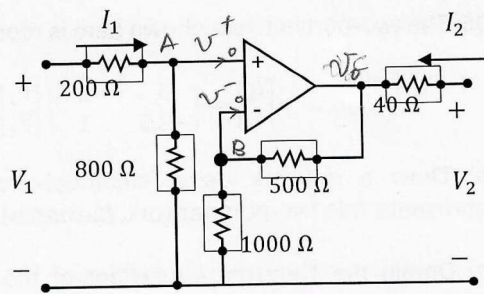
$\therefore V_{th} = 2 \cdot I_1$ & $R_{th} = 1 \Omega$



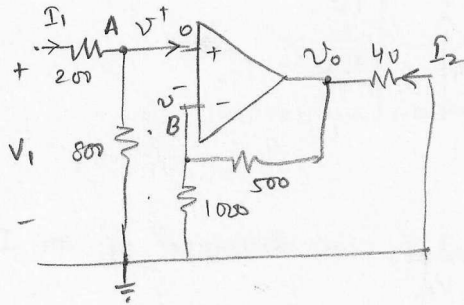
Q6. For the network shown in the figure, obtain the x -parameters that are defined as follows:

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

Assume that the Op-amp is ideal ($R_{in} = \infty$, $R_{out} = 0$, and $A_{gain} = \infty$). **(5-marks)**



if we set $V_2 = 0 \Rightarrow I_1 = x_{11} V_1$ & $I_2 = x_{21} V_1$



$$I_1 = \frac{V_1}{200 + 800} = \left(\frac{1}{1000}\right) \cdot V_1 \Rightarrow x_{11} = \frac{1}{1000} \Omega^{-1}$$

At node B.

$$V^- = \frac{1000}{1500} \cdot V_o = V^+ = \frac{800}{1000} \cdot V_1$$

$$\therefore \frac{1}{1.5} V_o = 0.8 V_1$$

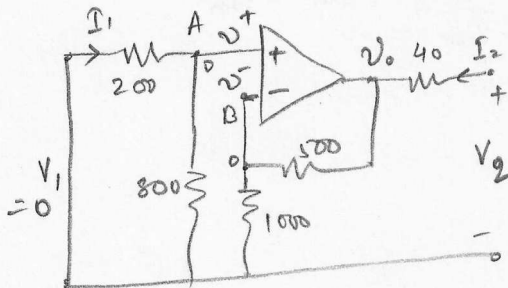
But $V_o = 0 - 40 I_2 = -40 I_2$

$$\therefore -\frac{40}{1.5} I_2 = 0.8 V_1 \Rightarrow I_2 = -0.03 V_1 \Rightarrow x_{21} = -0.03 \Omega^{-1}$$

2-marks for correct procedure

if we set $V_1 = 0$,

we get $I_1 = x_{12} V_2$
& $I_2 = x_{22} V_2$



$V_1 = 0 \Rightarrow I_1 = 0$ Since $V^+ = 0$ (ideal op-amp)

$$\therefore 0 = x_{12} \cdot V_2 \Rightarrow x_{12} = 0 \mid V_2 \neq 0$$

$$V^+ = 0 = V^- = \frac{1000}{1500} \cdot V_o \Rightarrow V_o = 0$$

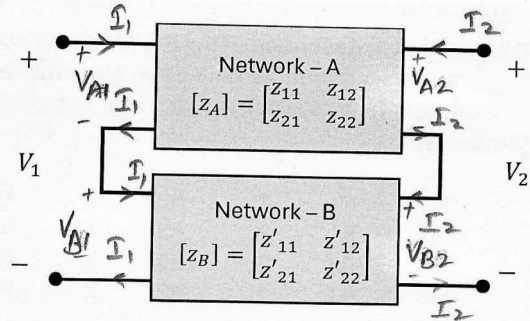
$$\therefore V_2 = 40 I_2 \text{ or } I_2 = \frac{1}{40} V_2 \Rightarrow x_{22} = \frac{1}{40} \Omega^{-1}$$

$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 0.001 & 0 \\ -0.03 & 0.025 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

3-marks for correct values of x -parameters. Deduct 1 mark for each incorrect entry.

Q7: Determine the equivalent representation of the network shown here which consists of two two-port networks connected in series. These networks are represented by their z - parameters.

(4-marks)



Network A :

$$\begin{bmatrix} V_{A1} \\ V_{A2} \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

Network B

$$\begin{bmatrix} V_{B1} \\ V_{B2} \end{bmatrix} = \begin{bmatrix} z'_{11} & z'_{12} \\ z'_{21} & z'_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix}$$

2-marks for correct procedure

$$\begin{aligned} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} &= \begin{bmatrix} V_{A1} + V_{B1} \\ V_{A2} + V_{B2} \end{bmatrix} = \begin{bmatrix} V_{A1} \\ V_{A2} \end{bmatrix} + \begin{bmatrix} V_{B1} \\ V_{B2} \end{bmatrix} = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} + \begin{bmatrix} z'_{11} & z'_{12} \\ z'_{21} & z'_{22} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \\ &= \left\{ \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{bmatrix} + \begin{bmatrix} z'_{11} & z'_{12} \\ z'_{21} & z'_{22} \end{bmatrix} \right\} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \\ &= \begin{bmatrix} z_{eq} \end{bmatrix} \cdot \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} \end{aligned}$$

$$\therefore [Z_{eq}] = [Z_A] + [Z_B]$$

2-marks for this final relation

Q8: A circuit is described using the following wire list, where each line represents a node and the elements connected to it. Each element has two terminals labeled 1 and 2. Assume that the reference node is N_0 , and the element direction is from terminal 1 $\rightarrow$ terminal 2 for every element.

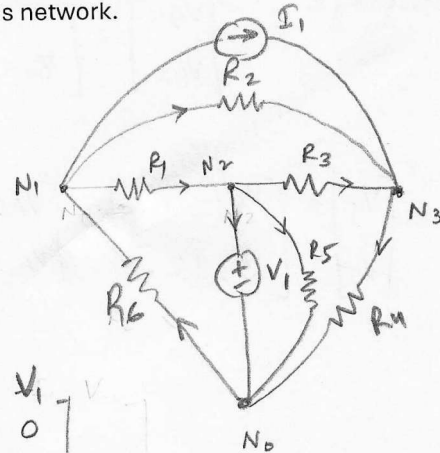
(5-marks)

*N1	$\checkmark$ R1	1	$\checkmark$ R2	1	$\checkmark$ I1	1	$\checkmark$ R6	2
*N2	$\checkmark$ R1	2	$\checkmark$ R3	1	$\checkmark$ V1	1	$\checkmark$ R5	1
*N3	$\checkmark$ R2	2	$\checkmark$ R3	2	R4	1	$\checkmark$ I1	2
*N0	R4	2	$\checkmark$ R5	2	$\checkmark$ R6	1	$\checkmark$ V1	2

The components in the network are represented as follows: Resistors $\rightarrow$ R1, R2, R3, R4, R5, R6; independent voltage source $\rightarrow$ V1; independent current source $\rightarrow$ I1.

- (a) Determine the total number of nodes and branches in the network. Please use the convention followed for the generalized-branch of an electrical network.
 (b) Construct the complete node-incidence matrix $[A]$ for this network.

(a) Total # of nodes : 4 1-mark
 Total # of Branches: 7 1-mark

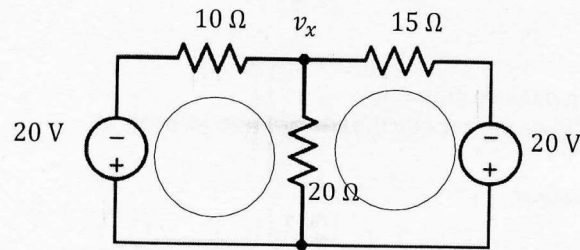


(b)

$$[A] = \begin{matrix} & \begin{matrix} R_1 & R_2 & R_3 & R_4 & R_5 & R_6 & V_1 \end{matrix} \\ \begin{matrix} N_1 \\ N_2 \\ N_3 \\ N_0 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & -1 & 0 \\ -1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & -1 & -1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & -1 & 1 & -1 \end{bmatrix} \end{matrix}$$

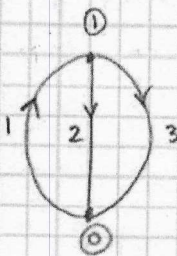
3-marks for first three rows. Deduct 0.5 mark for each incorrect entry. Rows may be interchanged, check the entries corresponding to elements. Deduct 1-mark if element numbers/names are not represented.

Q9: For the following circuit, obtain the value of v_x using graph-based approach. (7-marks)



Since node-voltage is required, we can use Node-Analysis in Graph-Approach
 i.e. $[v_n] = [Y]^{-1} \{ [A][i_s] + [A][Y_c][v_s] \}$, $[Y] = [A][Y_c][A]^T$

Graph:



$$[A] = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 1 & 1 \end{bmatrix}$$

1x3

1-mark. Deduct 0.5 for each incorrect entry

$$[Y_c] = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3x3

1-mark. Deduct 0.5 for each incorrect entry

$$[i_s] = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

3x1

1-mark. Deduct 0.5 for each incorrect entry

$$[v_s] = \begin{bmatrix} 1 & 2 & 3 \\ 20 & 0 & -20 \end{bmatrix}$$

3x1

1-mark. Deduct 0.5 for each incorrect entry

$$[Y] = [A][Y_c][A]^T = \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1/10 \\ 1/20 \\ 1/15 \end{bmatrix} = \frac{1}{10} + \frac{1}{20} + \frac{1}{15}$$

$$[Y] = 0.2167$$

2-marks for rest of the procedure.

$$\therefore [v_n] = [v_0] = [Y]^{-1} \{ [A][i_s] + [A][Y_c][v_s] \} = \left(\frac{1}{0.2167} \right) \left\{ \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 0 \\ -20 \end{bmatrix} \right\}$$

$$= \frac{1}{0.2167} \left\{ 0 - \frac{20}{10} - \frac{20}{15} \right\} = -15.38 \text{ V}$$

1-mark for correct final answer.

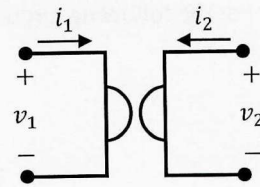
Q10: An ideal gyrator is shown here. The terminal characteristics are given as follows:

$$v_1 = -s i_2$$

$$v_2 = s i_1$$

(a) Determine if a gyrator is a passive element.

(b) Determine if a gyrator follows reciprocity theorem or not. (4-marks)



net power absorbed

$$(a) = v_1 i_1 + v_2 i_2 = v_1 i_1 + (s i_1) \left(-\frac{v_1}{s} \right)$$

$$= v_1 i_1 - v_1 i_1 = 0 \quad \therefore \text{passive element}$$

2-marks for showing this.

Do not give marks for conclusion is explanation is not given.

(b)

For reciprocity we need to check if.

$$\left. \frac{v_2}{i_1} \right|_{i_2=0} = \left. \frac{v_1}{i_2} \right|_{i_1=0} \quad \text{OR} \quad \left. \frac{i_2}{v_1} \right|_{v_2=0} = \left. \frac{i_1}{v_2} \right|_{v_1=0}$$

for $i_2=0$, $\frac{v_2}{i_1} = \frac{s i_1}{i_1} = s$

for $i_1=0$, $\frac{v_1}{i_2} = \frac{-s i_2}{i_2} = -s \neq \left. \frac{v_2}{i_1} \right|_{i_2=0} \Rightarrow \text{Not-reciprocal element}$

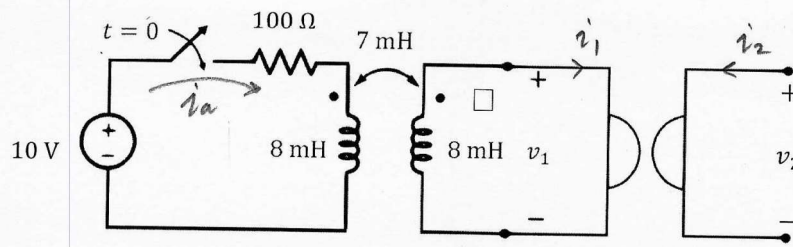
for $v_2=0$, $\left. \frac{i_2}{v_1} \right|_{v_2=0} = \frac{i_2}{-s i_2} = -\frac{1}{s}$

2-marks for showing either of these two.

for $v_1=0$, $\left. \frac{i_1}{v_2} \right|_{v_1=0} = \frac{i_1}{s i_1} = \frac{1}{s} \neq \left. \frac{i_2}{v_1} \right|_{v_2=0}$

Do not give marks for conclusion is explanation is not given.

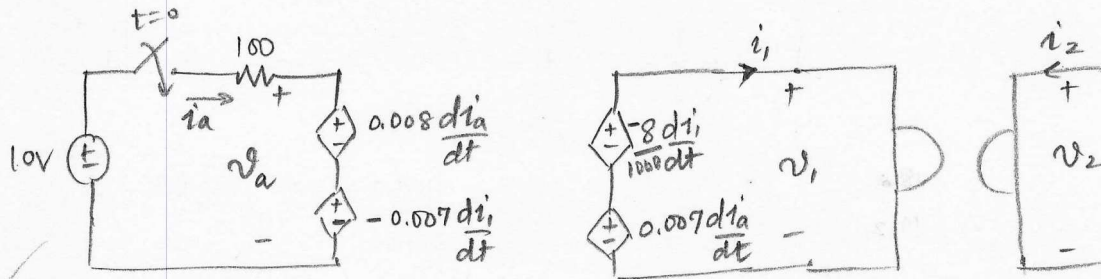
Q11: For the given network, the switch is moved to closed position at $t = 0$. Assume the value of gyrator-resistance as $s = 1$. Obtain the expression for v_2 valid for $t \geq 0$. Sketch the waveform of v_2 w.r.t. time. (6-marks)



$$v_1 = -s i_2, s=1 \Rightarrow v_1 = -i_2$$

$$v_2 = s i_1$$

$$\underline{v_2 = i_1}$$



Since $i_2 = 0$, (No-load connected) $\rightarrow v_1 = 0$.

KVL

$$10 - 100 i_a - 0.008 \frac{di_a}{dt} + 0.007 \frac{di_1}{dt} = 0 \quad \text{--- (A)}$$

$$0.007 \frac{di_a}{dt} - 0.008 \frac{di_1}{dt} = 0 \Rightarrow 0.008 \frac{di_1}{dt} = 0.007 \frac{di_a}{dt} \quad \text{--- (B)}$$

1-mark for one correct KVL/KCL expression.

$$10 - 100 i_a - 0.008 \frac{di_a}{dt} + 0.007 \cdot \frac{0.007}{0.008} \frac{di_a}{dt} = 0 \quad \therefore 10 - 100 i_a - (0.001875) \frac{di_a}{dt} = 0$$

1-mark for one correct KVL/KCL expression.

$$0.001875 \frac{di_a}{dt} + 100 i_a = 10 \approx \left(L \frac{di}{dt} + R i = V \right)$$

$$\therefore i_a = \frac{10}{100} (1 - e^{-t/\tau}), \tau = L/R = 18.75 \mu s.$$

2-marks for rest of the procedure.

Integrate $\frac{di_1}{dt} = 0.875 \frac{di_a}{dt} = (0.875) \cdot (0.1) \left(\frac{1}{\tau} \right) \cdot e^{-t/\tau} = 4666.67 e^{-t/\tau}$

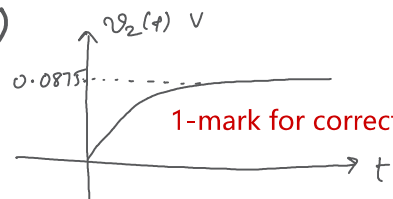
$$\therefore i_1 = 4666.67 (-\tau) e^{-t/\tau} + C = -0.0875 e^{-t/\tau} + C$$

$$i_1(t=0) = 0 \Rightarrow \cancel{-0.0875 e^{-0/\tau} = -0.0875}$$

$$0 = -0.0875 + C$$

$$i_1(t) = 0.0875 (1 - e^{-t/\tau}) = v_2(t)$$

1-mark for correct final expression



1-mark for correct plot